

A Survey on Explainable AI Narratives based on Large Language Models

MATTEO SILVESTRI, Sapienza University of Rome, Italy

VITTORIA VINEIS, Sapienza University of Rome, Italy

EDOARDO GABRIELLI, Sapienza University of Rome, Italy

FLAVIO GIORGI, Sapienza University of Rome, Italy

FABIANO VEGLIANTI, Sapienza University of Rome, Italy

FABRIZIO SILVESTRI, Sapienza University of Rome, Italy

GABRIELE TOLOMEI, Sapienza University of Rome, Italy

Explainable Artificial Intelligence (XAI) seeks to elucidate the inner logic of machine learning models, yet its outputs often remain difficult for non-technical users to understand. The emerging paradigm of XAI Narratives leverages Large Language Models (LLMs) to translate technical explanations into coherent, human-readable accounts. This survey provides the first systematic review of this approach, focusing on systems in which LLMs act as post-hoc narrative translators rather than autonomous explainers. We formalize this task as the Narrative Generation Problem, examine its integration with classical XAI methods such as feature attribution and counterfactual explanations across multiple data modalities, and introduce a taxonomy for narrative evaluation spanning three core dimensions. Finally, we analyze prompting strategies and outline open challenges and future directions for advancing reliable, interpretable, and context-aware XAI Narratives.

1 Introduction and Motivation

In recent years, machine learning (ML) has become deeply embedded in countless aspects of modern society, shaping decisions in domains as diverse as personalized recommendation systems, credit scoring, healthcare diagnostics, and judicial risk assessment [19]. This pervasive integration has amplified concerns about transparency, fairness, and accountability, prompting both academic and legislative attention. Notably, the European Union’s General Data Protection Regulation (GDPR) [96] has articulated a right to explanation for individuals affected by automated decisions. Additionally, the EU Artificial Intelligence Act [71] introduces concrete requirements for the transparency and interpretability of high-risk AI systems. These developments underscore an urgent need for Artificial Intelligence (AI) systems that are not only performant but also understandable, trustworthy, and aligned with human values.

In response to these imperatives, the field of Explainable Artificial Intelligence (XAI) has flourished, producing a vast array of methods to elucidate the internal mechanisms and decision-making rationales of opaque models. Classical XAI techniques—such as feature attribution, saliency mapping, rule extraction, and counterfactual reasoning (see, e.g., [33, 81, 87])—seek to expose the logic underlying a model’s predictions. However, despite their algorithmic sophistication, the outputs of these explainers are often intricate, mathematical, and complex for non-technical stakeholders to interpret. This mismatch between technical transparency and human interpretability remains a central challenge to the practical adoption of explainable AI.

To address this gap, researchers have increasingly turned to Large Language Models (LLMs) as a bridge between machine explanations and human understanding. By leveraging their generative and contextual reasoning abilities, LLMs can translate abstract or low-level explanatory signals into coherent, fluent, and audience-adapted narratives [16,

Authors’ Contact Information: Matteo Silvestri, silvestri.m@di.uniroma1.it, Sapienza University of Rome, Rome, Italy; Vittoria Vineis, vineis@diag.uniroma1.it, Sapienza University of Rome, Rome, Italy; Edoardo Gabrielli, edoardo.gabrielli@uniroma1.it, Sapienza University of Rome, Rome, Italy; Flavio Giorgi, giorgi@di.uniroma1.it, Sapienza University of Rome, Rome, Italy; Fabiano Veglianti, veglianti@diag.uniroma1.it, Sapienza University of Rome, Rome, Italy; Fabrizio Silvestri, Sapienza University of Rome, Rome, Italy; Gabriele Tolomei, Sapienza University of Rome, Rome, Italy.

31, 42, 64]. This emerging synergy between XAI and LLMs gives rise to a novel paradigm of *XAI Narratives*, consisting in the generation of *linguistically grounded, cognitively resonant natural language explanations*. This paper provides the first comprehensive, systematic survey dedicated exclusively to this paradigm, focusing on systems in which LLMs act as *post-hoc narrative translators* for the outputs of established explainers, distinguishing them from systems in which LLMs operate as explainers themselves. By systematically reviewing this literature, we aim to clarify conceptual boundaries between explanation and narrative, analyze the methodological landscape, and chart open challenges for the community. Beyond surveying technical approaches, we examine the evaluative aspects of XAI Narratives, focusing on their linguistic, cognitive, and practical dimensions. To structure this analysis, we propose a novel taxonomy for narrative evaluation, analyze the critical role of prompting strategies, and discuss the open challenges and future directions for this nascent paradigm. To foster transparency, reproducibility, and community collaboration, we accompany this survey with an open-access repository that aggregates all reviewed works, datasets, and evaluation resources: <https://github.com/hercolelab/awesome-explainable-AI-narratives>. We envision this resource as a living benchmark for future research on LLM-driven explainability.

The remainder of this survey is organized as follows. First, we position our contribution within the landscape of prior surveys and position papers on XAI and LLMs (Section 1.1). We then discuss the principal limitations, biases, and methodological challenges inherent to using LLMs for explainability, which motivate the development of XAI Narratives grounded in established explainers’ outputs (Section 1.2). Section 2 formalises the problem setting of XAI Narratives, distinguishing it from conventional explainability paradigms. Section 3 outlines our research questions and methods, including the search strategy, inclusion criteria, and the data curation process used to identify relevant works. Section 4 presents a comprehensive review of existing studies, systematically categorizing them according to the type of data they address and analyzing how LLMs have been employed to generate or translate model explanations into natural language narratives. Section 5 introduces a taxonomy of evaluation dimensions and methods for assessing the quality of narratives, and discusses emerging trends and current challenges. Section 6 explores the critical role of prompting strategies in shaping narrative coherence and factual grounding. Finally, Section 7 concludes the paper by summarizing the key insights and outlining open research directions and emerging opportunities for advancing this nascent paradigm for transparent and human-centered AI systems.

1.1 Comparison with Previous Work

Recent surveys and position papers have begun to explore the evolving intersection between XAI and LLMs, each emphasizing distinct aspects of this rapidly emerging research frontier. While these works have significantly advanced our collective understanding of how LLMs may contribute to interpretability, they often differ in scope, taxonomy, and conceptual focus, leading to a fragmented view of the field.

An earlier survey by Cambria et al. [13] specifically investigated presentation methods in XAI, focusing on natural language explanations and reviewing 70 papers published up to 2021. They proposed a roadmap and a multi-criteria decision-making model to help select XAI methods based on their natural-language presentation capabilities. While valuable in highlighting the limited attention paid to natural language presentation at the time, concluding that only a "handful" of works considered or implemented such explanations, its scope predates the widespread adoption and specific application of modern LLMs that define the current paradigm. Their review primarily covers earlier Natural Language Generation (NLG) techniques, such as template-filling, rather than the recent focus on LLMs as dedicated narrative generators.

More recently, Bilal et al. [8] offers a broad categorization of approaches, including post-hoc explanations and human-centered narratives, along with discussions of evaluation metrics, benchmark datasets, and applications. However, the work conflates methods in which LLMs generate explanations directly from raw data with those in which they act as translators, transforming existing explanations into user-centered narratives. Moreover, regarding this latter direction, the survey largely overlooks the literature on translation-based approaches, leaving a significant gap that we aim to fill comprehensively with this work. As we will argue in the Section 1.2, directly relying on LLMs even with contextual prompts is not sufficient, since the nature of explanation generation goes beyond pure reasoning abilities; instead, it requires grounding explainer outputs into well-structured, trustworthy narratives for end-users.

From a complementary yet critical perspective, Suh et al. [89] question the very premise of using LLMs to transform explainer outputs into natural-language narratives. They argue that such translations can inadvertently obscure inconsistencies, downplay uncertainty, and prioritize fluency over faithfulness—thus fostering an illusion of interpretability and encouraging user over-reliance. To mitigate these risks, they advocate for repurposing LLMs as “devil’s advocates” that probe explanations by highlighting uncertainty, alternative rationales, and latent biases, promoting critical rather than passive engagement. While this work offers a thought-provoking critique of the XAI Narratives paradigm—a discussion to which we return in Section 1.2—it does not provide a systematic overview or taxonomy of existing methodologies.

In contrast, our survey constitutes the first comprehensive synthesis dedicated exclusively to approaches that employ LLMs as post-hoc narrative generators — that is, systems that articulate, contextualize, and communicate the outputs of established explainers rather than functioning as explainers themselves. We systematically review and compare these narrative-generation frameworks across multiple domains, including tabular, image, graph, and policy tasks. Furthermore, we propose a unified taxonomy of evaluation metrics tailored to XAI Narratives that addresses the unique challenges of assessing factual grounding, linguistic quality, and user utility. Through this synthesis, we aim to establish the conceptual foundations and methodological standards necessary to advance this emerging paradigm of narrative-based explainability with LLMs.

1.2 On the Limitations of LLMs as Autonomous Explainers

Large Language Models have recently been repurposed for an expanding range of tasks beyond natural language understanding and generation [6], including domains traditionally reserved for model-specific explainability methods [8]. In XAI, LLMs have been employed to approximate or replace standard techniques for feature importance attribution, counterfactual reasoning, and even rule extraction. These approaches are motivated by the models’ general reasoning abilities and linguistic fluency, which allow them to produce seemingly coherent, human-like explanations without requiring access to the underlying predictive model. However, this autonomy comes at a cost. In this section, we argue that employing LLMs as standalone explainers is epistemically unreliable. Despite their narrative expressiveness, LLMs lack access to the internal decision mechanisms of the models they assert to explain, resulting in explanations that may be linguistically plausible but unfaithful to the actual computational process.

Recent work by Mayne et al. [65] provides compelling empirical evidence for this limitation. Their study systematically evaluated the capacity of state-of-the-art LLMs to generate valid and minimal counterfactual examples without any auxiliary model access or feedback. The results expose a fundamental epistemic gap: when prompted freely, LLMs tend to produce counterfactuals that are valid but excessively modified—altering numerous input features to flip a classifier’s prediction trivially. Conversely, when explicitly constrained to minimal edits, their counterfactuals often fail to alter the model’s decision. These findings suggest that LLMs, operating purely on linguistic priors, have no

fundamental understanding of the classifier’s decision boundary. They can articulate what might be changed, but not how much or why such a change would meaningfully affect the prediction. This undermines their suitability for causal or counterfactual reasoning tasks, which require precise sensitivity to a model’s decision landscape rather than surface-level associations.

In contrast to counterfactual reasoning, some studies report promising results when using LLMs to approximate feature importance attribution tasks. For instance, Mayne et al. [65] demonstrates that few-shot prompting with examples of feature–importance pairs enables an LLM to reproduce attribution patterns similar to those generated by classical XAI methods such as SHAP [61] or LIME [78]. While these findings indicate that LLMs can mimic attribution behavior under specific prompting conditions, they remain far from evidence of genuine interpretability. Two major caveats arise. First, the method relies on few-shot prompting with up to sixteen labeled examples—effectively requiring an external explainer to guide the model—thus defeating the claim of autonomous explanation. Second, the reported evaluations are conducted on canonical benchmark datasets such as Adult Income [5], Titanic [26], and Credit Card Default [105]. However, subsequent analyses [86] have shown that LLMs possess extensive prior knowledge of these widely available datasets, a phenomenon known as data contamination. As a result, the apparent performance gains may reflect memorization or statistical familiarity rather than genuine reasoning about the model’s behavior. Consequently, such results risk overestimating LLMs’ explanatory capabilities while masking potential biases embedded in their pretraining data.

Together, these findings underscore a central epistemological limitation: LLMs, when used as autonomous explainers, operate without direct access to the target model’s internal representations—its parameters, gradients, or intermediate activations. Unlike conventional XAI techniques that derive explanations from the model’s internal logic, LLM-based explanations rely entirely on linguistic priors and heuristic reasoning. This disconnect renders their outputs inherently indirect, potentially unfaithful, and susceptible to hallucination. Addressing this limitation motivates a new line of research focused on integrated XAI architectures, wherein LLMs are coupled with structured interfaces to a model’s internal states, enabling them to generate explanations that are both faithful and intelligible.

These limitations motivate our focus on XAI Narratives, where the LLM is not employed as an autonomous explainer but rather as a narrative translator, i.e., a system that transforms verifiable outputs from dedicated XAI modules into coherent, human-understandable accounts.

2 The Narrative Generation Problem in Explainable AI

The primary objective of XAI is to render the decision-making processes of complex machine learning models, often described as black boxes, comprehensible to human users [2, 102]. Traditional XAI techniques typically yield technical outputs, such as feature attribution scores, decision rules, or graph-based substructures, to make a model’s inner logic explicit. However, these outputs are often too abstract or mathematically dense for end-users without specialized expertise. Consequently, a growing body of research has sought to bridge this interpretability gap by translating technical explanations into natural-language narratives, enabling more accessible, contextually meaningful human understanding.

To formalize this process, we first define a technical explanation as the structured, model-derived artefact produced by an explainer. Let $f(\cdot)$ denote a predictive model and $\mathbf{x}$ an instance under analysis. An explainer function g maps the pair $(f, \mathbf{x})$ to a technical explanation $e_{tech} = g(f, \mathbf{x})$. Examples of e_{tech} include SHAP value vectors, rule sets derived from LIME, or explanatory subgraphs produced by GNNExplainer [106].

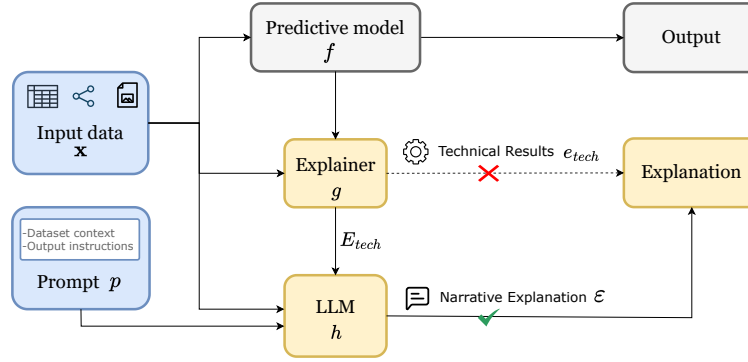


Fig. 1. *Illustration of the XAI Narrative framework.* A predictive model (f) processes input data ($\mathbf{x}$) to generate an output. Traditional explainers (g) provide only technical interpretations (e_{tech}). By incorporating contextual information (p) and leveraging an LLM, the system produces clearer, user-oriented explanations (ϵ), making model outcomes more understandable.

Building on this foundation, we conceptualize the *Narrative Generation Problem* (NGP) as the task of converting a technical explanation into a coherent, human-readable narrative. Specifically, given a technical explanation e_{tech} , a narrative generator h , and an optimal prompt p providing contextual and stylistic guidance, the goal is to produce a natural language narrative ϵ that faithfully and intelligibly conveys the content of e_{tech} (see Figure 1).

Definition 1 (The Narrative Generation Problem (NGP)). Given a predictive model f , an instance $\mathbf{x}$, and a corresponding technical explanation $e_{tech} = g(f, \mathbf{x})$, the objective is to derive a narrative explanation ϵ through a generative function h conditioned on a guiding prompt p :

$$\epsilon = h(p, \mathbf{x}, e_{tech}) \quad (1)$$

where h transforms the structured information about the instance $\mathbf{x}$ contained in e_{tech} into a linguistically coherent and semantically faithful textual narrative.

The function h is defined in a general sense to encompass a wide range of possible implementations, including deterministic mappings, probabilistic generative models, or hybrid architectures. In principle, h may be agnostic with respect to the underlying generative mechanism; however, this survey explicitly examines the role and impact of LLMs as the operative component of h , analyzing how they enable, structure, and potentially bias the generation of narratives.

A particular case of this formulation is the *counterfactual scenario*, in which the technical explanation e_{tech} is defined by the pair of factual and counterfactual instances [97] $(\mathbf{x}, \mathbf{x}')$. Therefore, the Counterfactual Narrative Generation Problem (CNGP) [31], corresponds to computing $\epsilon = h(p, \mathbf{x}, \mathbf{x}')$, where $\mathbf{x}' = \arg \min_{\mathbf{x}^*} d(\mathbf{x}, \mathbf{x}^*)$ s.t. $f(\mathbf{x}^*) \neq f(\mathbf{x})$.

It should be noted that the formulation in Definition 1 is intentionally general and designed to ensure applicability across diverse XAI settings. The definition remains deliberately agnostic to the distinction introduced by Cedro and Martens [16] between an *XAI description*, which enumerates factual associations, and an *XAI narrative*, which organizes such elements into a coherent, context-sensitive account. This abstraction allows ϵ to range from descriptive summaries to entirely narrative explanations, depending on the expressive capacity of h and the contextual guidance encoded in the prompt p . Furthermore, the prompt p serves as an explicit conditioning variable that controls communicative properties (style, context, and abstraction) without constraining its form. This treatment unifies both engineered and learned prompting strategies while enabling contextual enrichment through p , which can embed background knowledge,

user intent, or situational cues. This deliberate abstraction is precisely what allows our formulation to encapsulate the essential challenge at the core of XAI Narratives: aligning the formal rigor of algorithmic explanations with the communicative clarity of human narratives. It also provides a foundation for analyzing the diverse roles that LLMs can play in instantiating the function h , ranging from direct text generation to prompt-based translation of explainer outputs.

3 Research Questions and Methodology

To structure and guide this survey, we defined a set of research questions (RQs) designed to systematically investigate the role of LLMs in advancing XAI Narratives. Table 1 presents the thematic dimensions and their corresponding research questions that underpin our review. Collectively, these questions encompass the scope of XAI Narratives applications, the presence of conceptual frameworks, the nature of the datasets employed, the integration of LLMs within the explanation pipeline, and the evaluation methodologies used to assess the quality of the final outputs.

Table 1. Research questions guiding this survey.

Context	Research Questions
Foundations	RQ1: What are the main application domains of XAI Narratives? RQ2: What conceptual or architectural frameworks link LLMs with XAI? RQ3: How do LLMs complement, extend, or replace traditional explainers?
Data	RQ4: What types of data and explanation outputs are used for narrative generation? RQ5: How are datasets collected or adapted for LLM-based explanations?
Implementation	RQ6: What roles do LLMs play in producing narratives (e.g., translation, reasoning)? RQ7: How do prompting strategies and template designs influence narrative quality?
Evaluation	RQ8: How are narrative explanations evaluated in terms of quality, faithfulness, and usability?

The methodology adopted for this survey follows a systematic review protocol, structured into three main stages:

- (i) *Defining the scope and retrieving the literature:* We first delineated the survey’s scope by targeting studies published between 2017 and 2025 that investigate LLMs, XAI, and narrative-based explanation approaches. A comprehensive literature search was conducted across leading digital libraries and indexing services using carefully designed keyword combinations and Boolean operators to capture relevant contributions.
- (ii) *Screening and selecting studies:* The retrieved records were subjected to a multi-stage screening process involving duplicate removal, title and abstract assessment, and full-text evaluation. Inclusion criteria focused on studies that explicitly connect LLMs with explainability, interpretability, or narrative generation. Works lacking a clear explanatory or communicative component were excluded to maintain conceptual coherence.
- (iii) *Synthesizing and organizing the corpus:* The final corpus of selected studies was analyzed to extract recurring themes, methodological patterns, and research gaps. Each work was categorized by data type and thematic focus, providing the basis for the taxonomy and discussions presented in subsequent sections.

The overall objective of this systematic review is to produce a comprehensive and reproducible synthesis of current research at the intersection of LLMs and explainable AI. To ensure methodological robustness, we adhere to the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses* (PRISMA) guidelines [68], which provide a standardized framework for conducting and reporting systematic reviews transparently in the scientific literature.

Search Process. To ensure comprehensive coverage of the relevant literature, we conducted a systematic search from 2017 to 2025, focusing on the intersection of LLMs and XAI literature. The search was performed across major

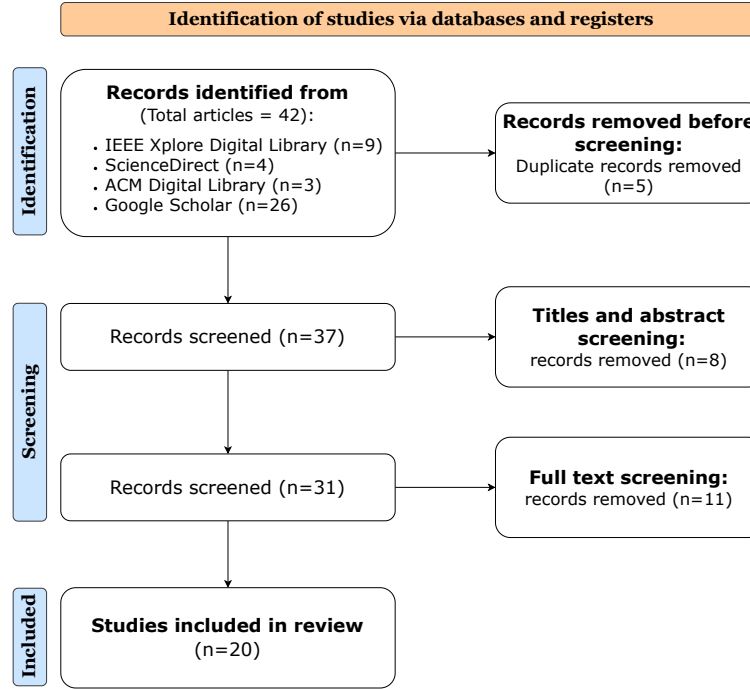


Fig. 2. Selection process of articles with the PRISMA flow diagram.

scholarly databases, including IEEE Xplore, ScienceDirect, ACM Digital Library, and Google Scholar. Advanced search functionalities and Boolean operators (e.g., AND, OR) were employed to maximize retrieval accuracy and scope. The search queries were carefully formulated to capture diverse strands of research at the convergence of explainability and natural language generation. Specifically, we used the following representative terms across titles, abstracts, and keywords: *"Explainable AI Narratives"*, *"Natural Language Explanations"*, *"Counterfactual Narratives"*, and *"Explanations with Large Language Models"*. Studies that did not align with the aims of this survey, such as those lacking a clear connection to LLMs, explainability, or narrative-based approaches, were systematically excluded during the screening process. To ensure methodological transparency, the selection workflow and the number of studies retained at each stage are summarized in the PRISMA flow diagram illustrated in Figure 2.

Inclusion/Exclusion Criteria. Conducting a literature search for a systematic survey poses significant challenges, particularly in a rapidly evolving research landscape such as that of LLMs and XAI. To ensure a focused and meaningful corpus, we established predefined inclusion and exclusion criteria to guide the screening process and maintain alignment with our research objectives. These criteria were instrumental in refining the selection to studies that directly address our research questions while minimizing the inclusion of tangential works. The inclusion criteria prioritized studies that: (i) integrate LLMs within an explainability pipeline, and (ii) employ LLMs specifically for the narrative generation stage, typically in combination with an existing explainer model (e.g., feature attribution methods, counterfactual explainers). Conversely, we excluded studies in which the LLM acts as the sole explainer without an explicit complementary

explanation mechanism. This distinction ensures that our analysis focuses on works where LLMs serve a communicative or translational role, rather than a purely inferential one.

In total, 42 articles were retrieved from the selected databases. After removing duplicates, 37 unique studies remained for screening. Title and abstract reviews led to the exclusion of 8 papers, followed by a full-text assessment that disqualified an additional 11 works. Ultimately, 20 articles met all inclusion criteria and were incorporated into the final corpus analyzed in this survey.

Data Extraction. The data extraction process was carefully designed to address the research questions guiding this survey systematically. Each of the 20 selected studies was examined in depth, and key information was extracted to enable comparative analysis and synthesis across the corpus. All data were compiled into a structured spreadsheet to ensure consistency and traceability throughout the review process. For each article, the following elements were systematically recorded: (i) bibliographic information, including title, publication year, and venue; (ii) dataset characteristics, specifying the type of data employed; (iii) explainer model integrated within the pipeline; (iv) role of LLMs within the explainability workflow; (v) evaluation metrics, primary findings, and reported limitations.

By systematically collecting and standardizing this information, we obtained a comprehensive overview of methodological choices, implementation strategies, and evaluation practices across the surveyed literature. This structured dataset forms the analytical foundation for identifying current trends, open challenges, and research gaps in the integration of large language models within explainable AI narratives.

4 Structured Overview of XAI Narratives Literature

In this section, we present a structured overview of the literature on XAI Narrative as per problem formulation in Section 2. Given the breadth of applications, including finance, medicine, molecular sciences, cybersecurity, and general machine learning tasks, as well as the variety of methodological approaches, we adopt a taxonomy that organizes relevant works according to the type of data at the center of analysis. Specifically, we distinguish contributions that focus on tabular data (Section 4.1), graph-structured data (Section 4.2), image data (Section 4.3), and reinforcement-learning policies (Section 4.4), as these modalities entail distinct representational challenges, explanation requirements, and narrative strategies. All these works are summarized in the following Table 2.

Table 2. *Summary of the studies reviewed in this survey.* For each work, the table reports the reference, the data modality (Tabular, Graphs, Images, and Policies), the specific LLMs and explainer methods employed (with citations to their respective original papers), and a concise synopsis (TL;DR) highlighting the main contribution.

Reference	Data Modality	LLMs	Explainers	TL;DR
Zytek et al. [113]	Tabular	GPT-4o [41]	SHAP [61]	Framework combining a Narrator for generating explanations and a Grader for automated evaluation.
Zeng [110]	Tabular	Mistral 7B [46]	SHAP[61]	Translate SHAP-based feature attributions into natural language explanations.
Castelnovo et al. [15]	Tabular	Mixtral 8x7B [47]	SHAP[61]	Conversion of SHAP explanations of banking models into narratives for Relationship Managers.

Table 2 – Continued from previous page

Reference	Data Modality	LLMs	Explainers	TL;DR
Burton et al. [12]	Tabular	BART [54], T5 [76]	SHAP [110], LIME [78], IG [90], LRP [9]	A pipeline that converts feature attributions into natural-language explanations using models fine-tuned on TEXEN, a corpus of expert-authored pairs linking explanations with their textual counterparts.
Swamy et al. [91]	Tabular	GPT-4o [41], Gemma2-9B [94], LLama3-70B [32]	LIME [78], CEM [24], MC-LIME [20]	A theory-grounded LLM-XAI framework that translates classical explainers into student-centred narratives through a multi-stage prompting pipeline.
Fredes and Vitria [27]	Tabular	GPT-4o [41]	DICE [67]	Human-like pipeline where an LLM extracts, verifies, and summarizes causes from sets of counterfactuals into user-friendly explanations
Giorgi et al. [31]	Tabular	DeepSeek [34], Qwen 2.5 [74]	DICE [67]	A two-stage pipeline that uses knowledge distillation to train small language models to generate and then refine counterfactual narratives.
Bhattacharya et al. [7]	Tabular	GPT-4-turbo [1]	DICE [67]	LLM-driven conversational framework that turns counterfactual explanations into interactive, user-friendly dialogues for healthcare
Mohanty et al. [66]	Tabular	Gemini 1.5 Pro [93]	NSGA-II [62]	A counterfactual XAI system for multi-disease prevention that integrates an LLM as an interactive interface.
Giorgi et al. [30]	Graphs	Qwen 2.5 [74]	CF-GNNExplainer [60]	It transforms graph counterfactual explanations into natural-language narratives, introducing new quantitative evaluation metrics (TNI, CCI)
Cedro and Martens [16]	Graphs	GPT-4o [41]	GNNExplainer [106]	GraphXAIN generates narrative explanations for graph models and formally distinguishes between descriptive and narrative XAI.
Pan et al. [69]	Graphs	GPT-4o [41], LLaMa3-8B [32], Gemma2-2B [94]	LRP, Saliency [9]	GraphNarrator trains a language model to transform saliency-based graph explanations into coherent narratives, using an iterative optimisation process.
Shirvani-Mahdavi and Li [84]	Graphs	GPT-3.5 [11], GPT-4o-mini [41], Gemini 2.0 Flash [92]	AMIE 3.5.1 [28]	It generates natural-language explanations for logical rules of knowledge graphs, combining variable type information with chain-of-thought prompting.
He et al. [37]	Graphs	GPT-3.5-Turbo [11], GPT-4 [1], BERT [23]	GNNExplainer [106], CF-GNNExplainer [60], CLEAR [63], RegExplainer [111]	LLM-GCE combines LLM-generated counterfactual text pairs with a graph autoencoder to produce chemically valid molecular counterfactuals.

Table 2 – Continued from previous page

Reference	Data Modality	LLMs	Explainers	TL;DR
Martens et al. [64]	Images, tabular	GPT-4 [1]	SHAP [61], SCAP [39]	CFStories uses LLMs to transform counterfactual visual explanations into concise narratives, demonstrating that structured prompts grounded in counterfactual regions yield clearer explanations.
Castellano et al. [14]	Images	LLaVa-NeXt [58], InstructBLIP [21], KOSMOS-2 [72]	Grad-CAM [83]	A multimodal framework that translates Grad-CAM visual explanations into textual narratives for classifying human vs. AI artworks.
Wojciechowski et al. [99]	Images	GPT-4 [1], Llama-3-70B [32]	LRP[3], MILAN[38]	Generate narratives that describe the image regions and neurons that contribute most significantly to the model’s classifications.
Yu et al. [107]	Images	Vicuna-7B [18], LLama3-1/3B [32], GPT-4 [1]	KFD [44], Grad-CAM [83]	LVLM-based deepfake detection framework that integrates multimodal feature alignment and LLMs to achieve generalizable and explainable detection.
Xi-Jia et al. [100]	Policies	GPT-3.5 [11], GPT-4 [1], LLama-2 [95], Starling-7B [112]	Dagger [79]	A model-agnostic framework that combines symbolic policy distillation with LLM reasoning to generate faithful, low-hallucination narratives of RL agent behaviour.
Kim et al. [50]	Policies	GPT-4.1 [1]	DeepSHAP [61]	Multi-agent LLM framework that unifies diverse XRL methods—and introduces new ones—within a natural language interface to deliver accurate, domain-aware insights into reinforcement learning agent behavior.

The surveyed works span diverse data modalities: ten on tabular data (50%), five on graphs, four on images ($\approx 20\%$), and two on reinforcement learning policies. Regarding model accessibility, fourteen employ closed-source LLMs (66%) and nine open-source ones. In terms of model families, eleven rely on OpenAI models ($> 50\%$), four on Google’s, five on LLaMA-based architectures, and seven on other frameworks (e.g., Qwen, DeepSeek, Vicuna). In the following sections, we also adopt specific terminology related to prompting strategies, as detailed in Section 6.

4.1 Tabular Data

This group includes all studies in which the underlying data are represented as feature vectors—a canonical format in many machine learning applications.

A representative contribution is that of Zyte et al. [113], which present Explingo, a framework that transforms conventional ML explanations—such as SHAP outputs—into coherent, human-readable narratives through the use of LLMs. The Explingo architecture consists of two complementary modules: the *Narrator*, which generates structured narratives via guided few-shot prompting, and the *Grader*, an automatic evaluation component that assesses the resulting explanations along multiple dimensions, including accuracy, completeness, fluency, and conciseness. The

authors demonstrate that the Grader’s assessments closely mirror human evaluations, suggesting its potential as both a scalable alternative to human annotators and a safeguard for quality assurance in deployment contexts. Empirical results indicate that, with carefully designed prompts and exemplar selection, LLMs can produce compelling narrative explanations. Nonetheless, the inclusion of exemplars can introduce additional complexity, potentially marginally reducing factual accuracy, underscoring the importance of fine-tuned prompt and exemplar design. Finally, integrating Explingo into an open-source toolkit highlights its applicability and relevance to practical XAI systems.

Building on this line of research, Zeng [110], who propose a framework that employs an LLM to translate SHAP-based explanations [61] for tabular models into natural language. Using the Titanic dataset as a case study in binary classification, the LLM is prompted to identify the most influential risk factors affecting survival. Subsequently, it generates a list of tuples that pair feature names with their corresponding SHAP values for a given instance. Within this setup, the language model effectively acts as a translator, transforming numerical attribution scores into human-readable narratives. Although the approach demonstrates the potential of LLMs to enhance interpretability, the study lacks a rigorous evaluation of explanation quality. Moreover, the authors explicitly acknowledge two central limitations: the sensitivity of results to prompt design and the occasional emergence of biased or misleading interpretations.

Extending this analysis, Castelnovo et al. [15] explore the use of LLMs to automate the generation of narrative explanations for AI-driven banking recommendations, thereby reducing the need for human intervention. Their framework employs an XGBoost model whose predictions are interpreted via SHAP, yielding feature-level attributions enriched with human-readable names, descriptions, and values. These outputs are then translated into domain-specific narratives through few-shot prompting with the Mixtral 8x7B model. Two prompting strategies are compared: *Approach U*, which leverages ten unrelated examples to encourage generalization, and *Approach C*, which clusters prompts around conceptually related variables curated by domain experts. The resulting narratives are benchmarked against a rule-based gold standard, which relies on extensive manual curation, using ROUGE-L to assess syntactic and cosine similarity to measure semantic alignment. Experimental results indicate that LLM-based explanations achieve performance comparable to handcrafted templates, offering scalable, business-oriented narratives that assist Relationship Managers in client-facing contexts. This study underscores the potential of LLMs to bridge the gap between technical explainability tools and human-centric communication in complex, high-stakes domains such as corporate banking.

Another milestone in the automation of textual explainability is the work of Burton et al. [12], who propose a complete pipeline for converting local-level numerical explanations into coherent natural-language narratives. Their framework extends the traditional XAI workflow—comprising classifier prediction followed by feature attribution (via LIME [78], SHAP, Integrated Gradients [90], or LRP [9])—with an additional Textual Explanation Generator that verbalizes structured explainer outputs. The pipeline is composed of three sequential modules: (i) an Explanation Processor that linearizes feature–importance pairs into placeholder-based text templates; (ii) a text generator (BART or T5) fine-tuned on a newly introduced corpus, TEXEN, which contains 496 expert-authored pairs of numerical and textual explanations; and (iii) a Post-Processor that restores the original feature and class names within the generated narrative. This architecture enables an end-to-end transformation of quantitative XAI outputs into human-readable accounts that maintain factual fidelity while enhancing interpretability. Experimental results on both the base and augmented TEXEN datasets demonstrate that fine-tuned T5 models outperform baselines, achieving 91% accuracy in a question–answer probing task on feature attributions. By formalizing a reproducible numerical-to-text explainability pipeline, this work lays the methodological groundwork for XAI Narratives systems, where large language models act as translators between structured interpretability artefacts and human reasoning.

A particularly noteworthy contribution is that of Martens et al. [64], who introduce a framework designed to transform SHAP explanations of Random Forest predictions into concise, human-readable narratives. The pipeline begins by contextualizing the dataset and target variable, provides a brief explanation of SHAP, highlights the most influential features by absolute contribution, and then prompts GPT-4 to generate an eight-sentence narrative accompanied by a summary. The approach was evaluated across datasets from sports analytics, education, and credit scoring. User studies involving both non-experts and data scientists revealed that over 90% of non-expert participants found the narratives convincing, intuitive, and helpful for decision-making. In contrast, approximately 78% of data scientists considered them valuable complements to traditional SHAP plots. Moreover, a controlled comprehension experiment demonstrated that participants achieved substantially higher accuracy when interpreting SHAPStories than when interpreting raw SHAP visualisations (e.g., 83.8% vs. 33.8% correct responses on a key question). Although some respondents expressed concerns regarding potential oversimplification or implicit assumptions, qualitative feedback consistently indicated that SHAPStories enhance the accessibility, persuasiveness, and practical usability of tabular model explanations.

More recently, Swamy et al. [91] present iLLuMinaTE, a comprehensive LLM-XAI framework that integrates insights from cognitive and social science theories of explanation to generate actionable, student-centered narratives in educational settings. The system operates as a zero-shot, chain-of-prompts pipeline comprising three stages—causal connection, explanation selection, and explanation presentation—which collectively map classical post-hoc explainers (e.g., LIME [78], CEM [24], and MC-LIME [20]) into pedagogically grounded, human-understandable feedback through large language models such as GPT-4o, Gemma2-9B, and Llama3-70B. Distinctively, iLLuMinaTE anchors its narrative generation in eight foundational theories of explanation, including the abnormal conditions model, Pearl’s causal framework, and contrastive reasoning. This theoretical grounding enables a principled transition from numerical feature attributions to meaning-rich, context-aware narratives. A large-scale evaluation across three MOOCs (21,915 generated explanations) and a user study involving 114 university students revealed that learners preferred iLLuMinaTE explanations in 89.5% of cases compared to traditional visual or textual explainers, citing superior usefulness, actionability, and trustworthiness. Overall, this work exemplifies a new generation of theory-aligned XAI Narratives systems, positioning LLMs as communicative mediators of explanatory intent rather than as autonomous explainers, and offering a framework for human-centered XAI across domains such as healthcare, finance, and social welfare.

Departing from earlier approaches, Fredes and Vitria [27] propose a multi-stage pipeline that transforms sets of counterfactuals (CFs) for tabular classifiers into coherent, user-oriented narrative explanations. The core idea is to guide an LLM through a sequence of human-like reasoning tasks: (i) generate CFs (e.g., using DiCE [67]) that reliably alter the model’s prediction; (ii) abduce candidate causes by prompting the LLM to compare the original instance with its CFs, deriving concise rules that capture the key changes responsible for the decision flip; (iii) verify and rank these candidate causes via program-aided prompting, in which the LLM writes and executes Python code to count how many CFs satisfy each rule, enabling support-based prioritization and mitigating hallucinations; and (iv) produce the final explanation, where the LLM, given the instance, CFs, dataset metadata, and rule supports, formulates a “call-to-action” narrative that highlights the most feasible and impactful modifications. The authors evaluate the pipeline across varying prompting strategies (zero-shot, one-shot, and tree-of-thought) and counterfactual set sizes (1, 3, and 5 CFs per instance). Results indicate that zero-shot prompting achieves the highest validity, with further improvement as CF sets grow larger. Tree-of-thought prompting enhances causal diversity but slightly reduces validity, whereas one-shot prompting underperforms overall. However, the evaluation relies primarily on LLM-based metrics (see Section 5), leaving aspects such as human-centered quality, causal soundness, and quantitative robustness underexplored. Overall, this study constitutes a pioneering attempt to test LLMs’ ability to reason over counterfactuals and translate

them into actionable, narrative guidance. It offers a promising direction for integrating language-based reasoning into counterfactual interpretability while underscoring the need for more rigorous evaluation frameworks.

Continuing along the trajectory of counterfactual reasoning, Giorgi et al. [31] propose the Multi-Narrative Refinement (MNR) pipeline, a novel framework for generating coherent, accessible counterfactual narratives, with a particular focus on tabular data. The approach employs a two-stage architecture that leverages knowledge distillation to transfer the reasoning capabilities of a large “teacher” model to a more efficient Small Language Model (SLM). In the first stage, a Draft Narrative Generator produces multiple candidate explanations for a given counterfactual instance. In the second stage, a Refiner model consolidates these drafts, resolving contradictions and integrating complementary insights to yield a single, more consistent and informative narrative. The authors demonstrate that this refinement step is essential, as it substantially enhances both factual accuracy and qualitative clarity, even enabling the SLM-based pipeline to surpass the performance of its larger teacher model. Beyond its conceptual innovation, MNR demonstrates substantial practical value by significantly reducing memory usage, inference time, and energy consumption—offering a scalable and resource-efficient solution for real-world XAI applications.

A complementary direction is explored by Bhattacharya et al. [7], who develop a pipeline that enhances counterfactual explanations through LLM-powered conversational agents, aiming to make them more actionable and comprehensible for non-expert users in healthcare contexts. The proposed system integrates counterfactual generation (via DiCE [67]) with a series of safeguards designed to ensure reliability and user trust: context-fusion prompting to ground LLM outputs, moderation filters to prevent harmful or irrelevant content, self-reflection through an LLM-as-a-judge mechanism to validate feasibility, and rule-based guardrails to mitigate counterintuitive recommendations. The resulting explanations are further enriched by constraining changes to actionable features and by providing interactive visualizations that enable data-centric what-if analyses. These design principles are instantiated in a chatbot for cardiovascular disease risk prediction, where users can explore hypothetical scenarios, receive step-by-step recommendations, and iteratively refine them through dialogue. A mixed-methods evaluation with 34 participants demonstrates the system’s effectiveness: most users completed justification, how-to, and what-if tasks; informed participants displayed deeper engagement and higher actionability; while novices—despite finding the interface intuitive—exhibited mild over-reliance on the model’s suggestions. Overall, the study shows that agent-augmented counterfactuals substantially enhance actionability, interpretability, and user satisfaction with minimal cognitive burden. However, it also exposes persistent challenges related to hallucination risks and the calibration of user trust in LLM-generated guidance.

In a closely related yet methodologically distinct approach, Mohanty et al. [66] introduce IMPACT, a system for multi-disease prevention that integrates counterfactual reasoning within an LLM-assisted XAI framework. The core component is a multi-dimensional counterfactual explanation model, optimized using NSGA-II [62], which identifies feature adjustments that jointly minimize the risk of diabetes and stroke. Unlike prior works where LLMs actively generate or structure the explanation, here the LLM (Google Gemini 1.5 Pro) plays a purely mediating role: it translates user queries into SQL operations for data retrieval. It reformulates the explainer’s outputs into conversational, user-friendly recommendations. In this sense, IMPACT embeds the LLM within the XAI pipeline not as an explanatory agent but as an interface layer—facilitating accessibility and engagement while preserving the autonomy and integrity of the underlying optimization-based explainer.

4.2 Graph-structured Data

Explanations over graph-structured data pose distinctive challenges compared to tabular settings owing to graphs’ inherently relational and non-Euclidean nature. In this domain, the explanatory task must capture not only the attributes

of individual nodes and edges but also the intricate dependencies encoded in the graph topology, where structural perturbations often carry semantic significance. As a result, generating faithful and interpretable narratives requires reasoning simultaneously about local features and global connectivity patterns.

One of the first studies to explore narrative explainability in graph-structured domains is that of Giorgi et al. [30], who address this challenge by introducing a novel framework that employs LLMs to generate natural-language narratives from technical counterfactual explanations on graphs. Their approach bridges existing state-of-the-art graph counterfactual explainers (see Table 2) with LLM-based narrative generation, prompting the model to translate formal counterfactual outputs into coherent, human-readable accounts. To systematically evaluate narrative quality, the authors propose two new quantitative metrics—Target Node Identification (TNI) and Counterfactual Class Identification (CCI)—that measure how accurately the LLM captures changes in node features and the neighbourhood structure, respectively. Experiments with open-source LLMs such as Qwen2.5 reveal two key findings: (i) explanation quality improves consistently with model scale, with the 14-billion-parameter variant achieving the best overall performance; and (ii) LLMs are substantially more effective at generating clear and actionable narratives when counterfactuals involve feature-level changes rather than structural modifications to the graph.

Building on this line of research, Cedro and Martens [16] present GraphXAIN, a framework that leverages GPT-4o to generate natural-language narratives for graph-based model explanations. A central conceptual contribution of their work is the formal distinction between an XAI Description—a static, context-free listing of salient features—and an XAI Narrative—a structured, story-like account that articulates cause-and-effect relationships within a given context. To assess the impact of this distinction, the authors conduct a comprehensive user study with machine learning researchers and practitioners, comparing three explanatory conditions: the raw output of GNNExplainer, the GraphXAIN-generated narrative, and a combined presentation featuring both. Results indicate that the narrative condition alone substantially improves understandability, user satisfaction, convincingness, and communicability, underscoring the added value of narrative structure in conveying graph-based explanations.

Another notable contribution is GraphNarrator by Pan et al. [69], which directly addresses the limitations of traditional saliency-based explainers that merely highlight important nodes without providing coherent textual reasoning. The framework trains a language model to map a Graph Neural Network (GNN) input–output pair into a structured narrative explanation. Specifically, it transforms saliency-based outputs into a Saliency Paragraph that encodes both the graph structure and feature importance, which then serves as the prompt for an LLM to generate an initial natural-language explanation. To iteratively improve the quality and fidelity of these pseudo-labels, the authors introduce a TAG Explanation Expert Iteration process. This feedback-driven training loop refines the pseudo-label generator by optimising for three objectives: faithfulness to essential inputs, alignment with the model’s predictions, and narrative conciseness.

The most recent contribution in this line of research is Rule2Text, introduced by Shirvani-Mahdavi and Li [84], a framework for generating natural-language explanations for logical rules extracted from knowledge graphs. In this approach, relational rules are first mined using the AMIE 3.5.1 algorithm on benchmark datasets, including FB15k [10], FB-CVT-REV, and FB+CVT-REV. Because such rules are often abstract and complex for non-experts to interpret, the authors systematically explore a variety of prompting strategies for large language models—covering zero-shot and few-shot setups, the inclusion of variable type information, and the use of chain-of-thought reasoning. Human evaluations assessing correctness, clarity, omission, and hallucination reveal that combining variable type cues with chain-of-thought prompting yields the most accurate and readable explanations. Experiments conducted with GPT-3.5 Turbo, GPT-4o Mini (restricted to chain-of-thought prompting), and Gemini 2.0 Flash (also with chain-of-thought

prompting) further demonstrate that explanation quality improves with both model scale and prompt sophistication. Finally, the authors examine the feasibility of using LLMs as automatic judges to replace human annotators, reporting moderate but consistent alignment between model-based and human evaluations.

Complementing the previous approaches, He et al. [37] adopt a distinct strategy oriented toward molecule-based tasks. Their framework, LLM-GCE, addresses a key limitation of existing graph counterfactual explainers—namely, the lack of domain-specific constraints, which often results in chemically infeasible or structurally unrealistic molecular graphs. To address this, the authors propose a multi-stage methodology that integrates linguistic reasoning with graph-based generation. In the first stage, rather than directly producing counterfactual graph structures, an LLM is prompted to generate counterfactual text pairs—high-level, human-interpretable descriptions that specify how a molecule should be modified to alter its predicted property. These textual instructions are then processed by a counterfactual autoencoder, which translates them into valid molecular graph topologies. A core innovation of the framework is a dynamic feedback module that mitigates hallucination risk by iteratively refining the counterfactual text pairs based on feedback from the autoencoder’s intermediate outputs and the downstream GNN’s classification results. In this setup, the natural-language component serves not only as a control signal for counterfactual generation but also as a narrative layer that implicitly explains the causal transformations underlying the model’s predictions.

4.3 Image Data

Image-based tasks remain among the most prominent and extensively studied domains in machine learning, and have likewise become a central focus in the field of explainable AI. Visual explanations, however, present unique challenges for narrative generation: the explanatory process must bridge perceptual features (e.g., object regions or textures) and semantic concepts comprehensible to human observers. Consequently, XAI Narratives in this modality aim not only to describe what visual elements drive a model’s decision, but also to articulate how alterations to those elements causally influence classification outcomes.

A representative contribution in this area is CFStories by Martens et al. [64], who employ GPT-4 to convert counterfactual explanations of image classifiers into natural-language narratives. Using SCAP explanations [39] with a Vision Transformer trained on the COCO dataset [57], the method identifies counterfactual regions by selectively blurring semantically meaningful image segments (e.g., blurring a “person” shifts the prediction from unicycle to skateboard). Prompts follow a consistent template: “The image was initially classified as [Initial Class]; however, when [Counterfactual Segment] was blurred, the model changed its classification to [Counterfactual Class].” In a large-scale user study with non-expert participants, 75% rated CFStories as equally or more convincing than their own manually written narratives, with all CFStories successfully incorporating counterfactual reasoning. Significantly, generation efficiency was markedly improved—participants required approximately 68 seconds to write a narrative, compared to only 5 seconds to read and assess a CFStory. The authors also note that LLM-generated narratives lacking access to counterfactual information often fail to capture misclassification dynamics, underscoring the need to integrate structured counterfactual signals into XAI Narratives for visual models.

A parallel contribution comes from Castellano et al. [14], who explore the use of multimodal large language models to generate natural-language explanations for model decisions in the task of distinguishing human-created from AI-generated artworks. Their framework integrates visual and textual modalities by combining Grad-CAM [83] heatmaps—overlaid on the original images to highlight salient visual regions—with carefully structured textual prompts that guide the LLM in translating these visual cues into coherent narrative explanations. Despite the conceptual appeal of this multimodal setup, quantitative evaluations reveal limited effectiveness: the generated explanations achieve

relatively low scores in image-to-text similarity (the degree to which the narrative aligns with the highlighted visual evidence) and text-to-label alignment (the degree to which the explanation supports the predicted class). These results suggest that, while multimodal LLMs hold promise for bridging visual and textual interpretability, substantial progress is still needed to ensure faithfulness and semantic coherence in narrative explanations for visual classification tasks.

Another significant contribution is presented by Wojciechowski et al. [99], who propose a pipeline for generating faithful and plausible natural-language explanations for CNN-based image classifiers. Their method first identifies influential neurons using Layer-wise Relevance Propagation (LRP) [9], then annotates them with concise semantic descriptions via MILAN [38], linking each neuron to its corresponding image region. The resulting meaning representation, stored in a structured JSON format, is subsequently provided as input to an LLM, which synthesizes it into a coherent natural-language explanation. In experiments on ImageNet using a ResNet-18 [36] backbone, the proposed approach outperformed established baselines such as Show, Attend and Tell [48, 101] and NLX-GPT [80] in terms of explanation faithfulness. Faithfulness was assessed through targeted intervention experiments: image regions were selectively masked based on the generated explanations, and influential neurons were ablated according to the MILAN-annotated relevance lists. The resulting class-flip rates and probability reductions quantitatively validated the alignment between the explanations and the model’s behavior. Human evaluations further confirmed that the generated narratives were judged substantially more plausible than those of competing methods, though slightly less comprehensible, with occasional hallucinations (8%) and frequent omissions of relevant details (44%).

In a recent and innovative advance, Yu et al. [107] introduce a Large Vision-Language Model (LVLM)-driven framework that redefines deepfake detection as a multimodal, explainable reasoning task rather than a purely visual classification problem. By fusing fine-grained visual-textual alignment with language-based interpretability, the approach leverages LLMs’ reasoning capabilities to produce both highly generalizable and human-understandable deepfake detection outcomes. The authors identify a key challenge: conventional deepfake detectors are limited by their reliance on handcrafted features or dataset-specific cues, while existing LVLMs, though powerful, lack sensitivity to subtle visual artifacts. To bridge this gap, the paper proposes a Knowledge-guided Forgery Detection (KFD) [44] module coupled with a Forgery Prompt Learner (FPL) that together interface with an LLM to generate interpretable detection narratives. The KFD aligns multimodal features by correlating image embeddings with textual embeddings describing pristine and manipulated faces (e.g., “inconsistent head poses,” “unnatural blending edges”), many of which are generated via GPT-4 and curated by humans. These correlations yield “forgery consistency maps,” which guide forgery localization and classification. The FPL then converts these outputs into fine-grained forgery prompt embeddings, which—together with visual and question prompts—are passed to the LLM (Vicuna-7B) for response generation. The system is trained using simulated forgery-text pairs that mimic deepfake artifacts and evaluated across multiple benchmarks. Results show superior AUC and AP scores compared to both traditional and LVLM-based baselines, along with robust performance across unseen manipulation types and image perturbations. Crucially, the model also supports multi-turn dialogue, enabling users to query the reasoning behind detection outcomes—thus positioning LVLMs as explainable forensic agents.

4.4 Reinforcement-Learning Policies

Policy-based tasks are a cornerstone of reinforcement learning (RL) and sequential decision-making research, in which agents learn to optimize behavior through continuous interaction with dynamic environments. As these systems are increasingly deployed in domains such as robotics, autonomous driving, and adaptive recommendation, the need to make their decision-making processes transparent and interpretable has become paramount. Explaining why an agent

acts in a certain way, and how its learned policy generalizes across states, is central not only to ensuring accountability and safety but also to fostering human trust and collaboration. Within this context, XAI Narratives for RL has emerged as a promising avenue—one that integrates symbolic, statistical, and linguistic reasoning to translate abstract policy representations into human-understandable explanations of behavior.

A pioneering contribution in this area is made by Xi-Jia et al. [100], who introduce a model-agnostic, behavior-representation-driven framework that unites symbolic distillation with LLMs’ reasoning to produce natural-language explanations of agent behavior. The method first extracts a structured, symbolic representation of a black-box policy, capturing decision boundaries and state–action dependencies in a form suitable for downstream reasoning. This distilled representation serves as the grounding context for an LLM, which is prompted to generate explanations describing the agent’s behavior, objectives, and deviations from optimality in various scenarios. Through this integration, the framework constrains the LLMs’ reasoning space, reducing hallucinations and ensuring that generated narratives remain faithful to the underlying policy dynamics. The authors validate their approach across multiple RL environments—including grid-based navigation and continuous-control tasks—and through human studies assessing the clarity, trustworthiness, and factual consistency of the generated explanations. Results demonstrate that the proposed method consistently outperforms purely text-based or black-box LLM explainers, producing narratives that are not only more accurate and faithful but also perceived by human evaluators as more insightful and trustworthy. By demonstrating that structured policy representations can effectively guide LLM reasoning, this work represents a significant step toward human-aligned, narrative explainability for autonomous agents, bridging the gap between symbolic interpretability and natural-language understanding.

Expanding the role of large language models in policy interpretability, Kim et al. [50] propose TalkToAgent. This human-centered framework integrates explainable reinforcement learning (XRL) with LLM-based multi-agent reasoning to generate multimodal, interactive explanations of agent behavior. Motivated by the limitations of existing XRL methods—fragmented interpretability, limited accessibility for non-experts, and restricted counterfactual reasoning—the authors design a modular architecture comprising five specialized agents: a Coordinator that interprets user queries and routes them to the appropriate XRL module; an explainer that produces domain-aware textual accounts of visual outputs; a Coder that autonomously generates Python code for decomposed reward functions or counterfactual policies; an Evaluator that verifies correctness and alignment with user intent; and a Debugger that iteratively refines the code through guided feedback loops. Implemented primarily with GPT-4.1, these agents collaborate to unify diverse XRL tasks—feature attribution, expected-outcome decomposition, and counterfactual reasoning—within a conversational interface. The system integrates classical interpretability methods, including DeepSHAP for feature importance, temporal and component-wise Q-value decomposition for expected outcomes, and three complementary counterfactual explanation modes: (i) action-based (altering specific actions), (ii) behavior-based (expressed via qualitative behavioral descriptors such as conservative or aggressive, parameterized by a smoothing factor α), and (iii) policy-based (yielding executable, rule-based control strategies). The framework is validated on the nonlinear quadruple-tank process control benchmark using a Soft Actor–Critic agent [35]. Results show high task-classification accuracy (up to 97.5%), consistent and reliable counterfactual code generation with minimal hallucination—achieved through coder–debugger interaction—and qualitatively robust explanations that successfully link RL decisions to domain semantics. These findings illustrate that LLMs can effectively interpret user intent, orchestrate XRL computation, simulate alternative scenarios, and communicate insights through natural language grounded in system dynamics. By reframing explanation as an interactive dialogue rather than a static visualization, TalkToAgent bridges the gap between opaque RL policies and human understanding, marking a significant advance toward narrative, intent-aligned, and human-centric explainable

reinforcement learning. The authors conclude by outlining future extensions, including goal-conditioned counterfactual generation, value-based policy analysis, and broader domain generalization—positioning TalkToAgent as a crucial step toward fully conversational XAI systems for autonomous decision-making.

5 Metrics and Evaluation

A central challenge in advancing XAI narratives is rigorously evaluating natural-language explanations. Unlike purely quantitative outputs, such explanations must be assessed across multiple dimensions—from their fidelity to the underlying model to their usefulness and comprehensibility for human users. Establishing a systematic evaluation framework is therefore essential to ensure both scientific validity and practical relevance. In this section, we introduce a taxonomy of evaluation dimensions and methods, situate the surveyed literature within this framework, and illustrate how different studies assess XAI narratives.

5.1 Taxonomy of Evaluation Dimensions and Methods

Coming at the crossroads of natural language generation (NLG) and explanation evaluation, the assessment of LLM-generated narratives inherits challenges from both domains. On the one hand, it must account for textual qualities such as fluency and clarity; on the other, it must ensure that explanations remain faithful to the underlying explainer and meaningful for human users. Moreover, unlike traditional explainer outputs, such as SHAP importance scores or saliency maps, natural language narratives introduce an additional layer of variability: their meaning is mediated not only by the underlying attribution signal but also by linguistic choices, style, and contextual framing, further complicating their evaluation. As a result, a diverse set of metrics and evaluation strategies has emerged in the literature.

Although Bilal et al. [8] focus on the evaluation of LLMs that directly generate model explanations, the evaluative distinction they propose remains broadly applicable to our setting, where LLMs verbalize the outputs of separate explainer models. Specifically, they distinguish between *qualitative, human-centered* approaches and *quantitative, model-centered* approaches. The former assess explanations in terms of *comprehensibility*, referring to the clarity with which the reasoning is communicated and the ease of understanding for human users, and *controllability*, which denotes the extent to which users can interact with and refine explanations through feedback or by focusing on specific parts of the decision process. Such evaluations are typically realized through user studies in which participants highlight unclear or incomplete elements and systems, and iteratively adapt their responses. By contrast, quantitative, model-centered approaches evaluate the epistemic reliability of explanations, focusing on *faithfulness*, that is, the degree to which an explanation accurately represents the model’s internal logic, causal structure, and learned dependencies, and *plausibility*, understood as the logical coherence and consistency of the explanation with domain knowledge. A similar distinction between human-centered and computer-centered XAI evaluations was already proposed by Lopes et al. [59], who propose a taxonomy that organizes evaluation methods into these two distinct families. In their framework, *human-centered* methods require user experiments with human subjects to assess key properties such as trust, understandability, performance, and the usefulness and satisfaction of explanations. By contrast, *computer-centered* methods rely on formal definitions and computational metrics to evaluate an explanation’s quality without human involvement, focusing on aspects like interpretability and fidelity. Building on a more purpose-focused perspective, Speith and Langer [88] propose classifying evaluation methods based on which part of the XAI process they target: the quality of the *explanatory information* itself (e.g., its fidelity and completeness), the resulting user *understanding* of the system, or the final impact on *desiderata* (e.g., trust, acceptance, and human-machine performance). Collectively, these frameworks highlight a fundamental dualism in XAI evaluation, distinguishing between the evaluation method (human-

vs. computer-centered) and its purpose (assessing the information, the resulting understanding, or the outcome). This dual perspective provides a sound conceptual foundation for our taxonomy, which extends these notions into three macro-dimensions: Faithfulness to the Explainer, Linguistic Realization, and Human Utility.

The foundational pillar of narrative evaluation is, of course, the explainer itself: fidelity, completeness, and stability of its outputs are necessary preconditions for generating meaningful natural language explanations. If the explainer is unfaithful or unstable, even the most linguistically fluent narrative cannot provide a reliable account of the model’s reasoning. This challenge is compounded by LLMs’ tendency to produce text that appears coherent and plausible regardless of the accuracy of the underlying signal or the true causal path, as noted by Ichmoukhamedov et al. [42]. LLMs often “self-correct” implausible statements to make them seem internally consistent, which can obscure inaccuracies in the explainer’s output. Nevertheless, since most studies implicitly assume the reliability of the explainer, we follow this convention and focus our analysis on the textualization stage—the transformation of explainer outputs into natural-language narratives. To this aim, we propose a fine-grained taxonomy that organizes evaluation criteria into three *evaluation dimensions*, each associated with a distinct overarching goal, and each encompassing specific criteria widely discussed in the NLG and XAI literature, namely:

1. *Faithfulness to the Explainer (FE)*, capturing the alignment of the narrative with the outputs of the underlying explainer model. When FE is satisfied, epistemic reliability is achieved. FE encompasses the following criteria:
 - (a) *Correctness* – i.e., factual agreement with the explainer’s outputs.
 - (b) *Completeness* – i.e., coverage of all relevant features or structural elements.
2. *Quality of Linguistic Realization (LR)*, focusing on the quality of the explanation as natural language text. When LR is satisfied, linguistic efficacy is achieved. LR encompasses the following criteria:
 - (a) *Appropriateness of terminology* – i.e., the use of domain-appropriate, precise, and accessible vocabulary that matches the target audience.
 - (b) *Fluency* – i.e., the grammatical correctness, readability, and natural flow of the explanation as text.
 - (c) *Efficiency of expression* – i.e., the ability of the explanation to convey all necessary information without redundancy or irrelevant elaboration. Unlike the narrower notion of conciseness, this criterion recognizes that the optimal length and complexity of an explanation vary by task and audience: explanations may justifiably be longer to ensure completeness and contextualization, or shorter to facilitate rapid understanding [40, 109]. Its assessment often requires an explicit judgment call by the XAI system designers.
3. *Human Utility (HU)*, reflecting the value of the explanation for end users. When HU is satisfied, pragmatic relevance is achieved. HU encompasses the following criteria:
 - (a) *Cognitive fit* – i.e., the degree to which the explanation is plausible given logic and context, understandable, and internally consistent.
 - (b) *Attitudinal response* – i.e., the extent to which the explanation fosters user trust, confidence, and satisfaction.
 - (c) *Use value* – i.e., the practical benefits of the explanation, such as actionability and insightfulness.

Complementing these content-focused dimensions (what to measure), we identify three families of *evaluation methods* (how to measure), spanning a continuum from quantitative to qualitative assessment:

1. *Automatic approaches* — quantitative methods that rely on formal metrics and can be divided into:
 - (a) *Reference-based*: compare generated narratives to human-written or gold-standard references using surface-level metrics such as BLEU [70], ROUGE [56], and METEOR [4], or semantic metrics such as BLEURT [82] and embedding-based cosine similarity.

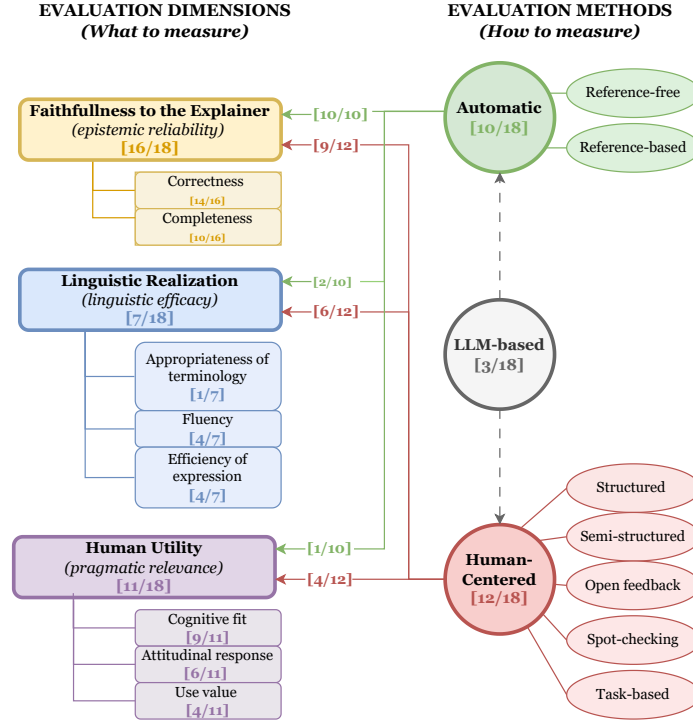


Fig. 3. *Proposed taxonomy for evaluating LLM-generated narratives.* Content dimensions (Faithfulness to the Explainer, Linguistic Realization, and Human Utility) and their underlying criteria define *what* to measure. In contrast, evaluation methods (Automatic, LLM-based, Human-centered) define *how* to measure. Numbers in brackets indicate the proportion of analyzed papers that address each dimension, criterion, or methodological family relative to the total number of studies in the corresponding category.

- (b) *Reference-free*: evaluate intrinsic properties of the narratives—such as length, perplexity [43], or entropy—or measure alignment with the explainer’s structured outputs (e.g., attributions, heatmaps).
2. *LLM-based assessments* — where LLMs act as evaluators rather than natural language translators of the explanations. Two main configurations are observed:
 - (a) *LLM-as-Judge*, when models assign rubric-based or comparative scores, often benchmarked against human ratings for validation.
 - (b) *LLM-as-User*, when models emulate end-user judgments by rating qualities such as clarity, trust, or usability. Although conceptually distinct, much of the literature treats these settings interchangeably; for clarity, we use *LLM-based* as an umbrella label encompassing both.
3. *Human-centered approaches* — qualitative or mixed methods to gather human input using tools like structured surveys (e.g., Likert scales, pairwise comparisons), semi-structured interviews, open-ended feedback, spot-checking, or task-based evaluation studies.

It is important to note that, while automatic methods are typically used to assess FE or LR, and human-centered methods are primarily used to capture HU aspects, several exceptions are reported in the literature. For example, objective measures like users’ response time can be integrated into task-based human evaluations. In contrast, human judgments are often used to assess dimensions typically associated with LR, such as text clarity.

Figure 3 provides a visual overview of the proposed evaluation taxonomy, illustrating how the three methodological families map onto the three macro-level evaluation dimensions. The relative frequencies indicated in brackets provide an empirical snapshot of current research practices, revealing that most existing studies prioritize epistemic reliability, particularly correctness-oriented measures of faithfulness. In contrast, considerably fewer works explicitly evaluate linguistic quality or user-centered outcomes. Notably, aspects of LR are often not assessed as independent constructs. Still, they are inferred indirectly through their downstream effects on HU—for example, by examining how fluency, style, or phrasing influence perceived clarity, trust, or task performance. This overlap underscores the interdependence of the three macro-dimensions. A closer inspection of the surveyed corpus further highlights the uneven methodological coverage across studies: only five out of eighteen papers address all three dimensions simultaneously [25, 31, 84, 99, 114], while another four combine human-centered evaluations with automatic metrics [14, 84, 99, 100]. Strikingly, just two studies integrate all dimensions *and* all methodological families, suggesting that comprehensive, multi-perspective evaluation remains the exception rather than the norm.

Rather than discussing each paper individually, the following sections synthesize prior work according to the criteria outlined in our taxonomy, analyzing how different studies instantiate, operationalize, and measure each evaluation dimension. Before examining how existing works instantiate the proposed evaluation dimensions, it is worth clarifying that four of the previously discussed studies [37, 49, 50, 110] are excluded, as they do not provide a systematic evaluation of the generated narratives.

Alternatively, the following discussion considers an additional set of papers that specifically focus on evaluating explanations generated by LLMs. In particular, these works outline a methodological trajectory for evaluating XAI narratives, from human-centered qualitative studies to automated, LLM-based, and hybrid evaluation frameworks. While Zytek et al. [114] ground their evaluation in direct human judgments of narrative soundness, completeness, and clarity, De Bona et al. [22] and Domnich et al. [25] progressively shift toward using LLMs as evaluators, either as surrogates for human participants or as fine-tuned predictors of human preferences (Section 5.2). In parallel, Ichmoukhamedov et al. [42] propose fully automatic, metric-driven evaluation methods that quantify faithfulness without human input. This evolution reflects a broader trend toward scalable, reproducible, and hybrid evaluation strategies for XAI narratives, balancing the interpretive reliability of human assessment with the efficiency of LLM-based evaluation.

5.1.1 Faithfulness to the Explainer

Faithfulness to the explainer captures how well a generated narrative reflects the attribution process of the underlying model. In our taxonomy, faithfulness unfolds along two criteria, namely *correctness* and *completeness*. In the literature, these criteria have been operationalized and measured through three complementary strategies: human evaluation, automatic metrics, and LLM-as-judge.

Correctness. *Human-centered evaluations* typically operationalize correctness as factual agreement with the explainer’s suggested changes or attributions. Giorgi et al. [30] asked human evaluators to rate, on a 5-point Likert scale, whether a narrative correctly described counterfactual changes in graph structures; in this case, correctness was defined as factual alignment with explainer-suggested changes. Similarly, Zytek et al. [114] evaluates the *soundness* of narratives

against SHAP scores through participant ratings on a 3-point anchored Likert scale¹. In another study, Zytek et al. [113] evaluated *accuracy* by fine-tuning an LLM to perform rubric-based scoring, achieving 96% alignment with human evaluators in a boolean setting.

However, evaluating faithfulness in user studies has also relied on more indirect methods. For instance, Martens et al. [64] measure correctness as the proportion of participants’ answers that align with the model’s ground-truth outputs, either selecting the correct option in comprehension tasks or matching the model’s decision in a loan approval scenario.

Fredes and Vitria [27] instead propose a “closed-loop” evaluation, where an LLM is asked to generate a counterfactual example using only the original case, dataset information, and the provided explanation. The successful generation of a valid counterfactual is interpreted as evidence that the explanation contained the essential information, thus operationalizing correctness through counterfactual reconstruction.

Quite apart from the rest of the literature for their focus on generating policy explanations in interactive, goal-directed environments, Xi-Jia et al. [100] propose a model-agnostic approach to assessing faithfulness: they construct annotated datasets by sampling agent behaviors from heuristic policies, which provide access to the agent’s internal beliefs and intents and enable the programmatic generation of ground-truth explanations for each observation. The resulting LLM-generated narratives are then automatically ranked by an LLM according to five criteria, namely informativeness, correctness, the identification of the agent’s strategy goal, and goal category. Interestingly, the authors also quantify hallucination frequency using both LLM- and human-based detection across generated explanations and their downstream action predictions. Results show that hallucinations are significantly reduced when narratives are grounded in decision-path representations compared to state-only or ungrounded prompts, indicating that explanations informed by structured decision traces produce narratives that are more faithful to the agent’s actual reasoning. This supports the view that explainer-informed narratives are more epistemically reliable than purely generative or observational ones. Moreover, statistical analyses reveal that users cannot reliably detect hallucinated content, underscoring a mismatch between perceived and actual correctness and motivating the need for intrinsic hallucination control in narrative generation.

In a symbolic, rule-based setting, Shirvani-Mahdavi and Li [84] assess whether LLM-generated narratives accurately verbalize the logical structure of rules extracted from knowledge graphs using AMIE (Association Rule Mining under Incomplete Evidence) [29]. In this work, correctness is rated by human annotators on a 5-point Likert scale, assessing whether all rule components are included in the exact logical order specified by the rule. To complement these human assessments, the authors introduce an LLM-as-a-judge framework validated against human ratings. The LLM reproduces human judgments with substantial rank-order alignment and moderate absolute agreement, indicating that it reliably distinguishes higher-quality explanations despite some variation in score calibration. Moreover, it consistently identifies imperfect narratives, supporting its value for automated evaluation. Complementary analyses further quantify hallucinated and missed entities and relations, offering a fine-grained measure of factual correctness relative to the symbolic ground truth.

Reference-based automatic approaches measure correctness against expert-authored gold narratives using similarity metrics. In this setting, the anchor for evaluation is not the explainer’s raw output but the human textualization of it. Correctness is thus captured only indirectly, under the assumption that expert-written narratives faithfully represent the explainer. As a consequence, faithfulness becomes partly dependent on expert perception and phrasing, with limitations related to potential subjectivity and inconsistency, as noted by Yao et al. [103]. In this setting, Castelnovo et al. [15]

¹Unlike scales that only provide numbers, anchored Likert scales include a descriptive sentence for each level. The goal of these descriptions is to reduce subjectivity.

relied on syntactic similarity (ROUGE-L) and semantic similarity (embedding cosine) to assess correctness relative to expert references. Burton et al. [12] adopted a broader metric suite, namely BLEU and METEOR for syntactic similarity, BLEURT for semantic similarity, and further conducted a manual error analysis of 30 narratives, categorizing errors by explanation type to identify subtle correctness failures and missing explanatory factors not detected by automatic scores. Ichmoukhamedov et al. [42] complemented their SHAP-based agreement metrics with a human-similarity dimension, comparing generated narratives against expert-written references using BLEURT, METEOR, and cosine similarity. Consistent with the findings of Shirvani-Mahdavi and Li [84], the authors report that modern embedding spaces are sufficiently expressive to distinguish between faithful and unfaithful narratives, even with simple cosine similarity. At the same time, they cautioned that these differences were often modest: stylistic variation could be mistaken for factual error, while manipulations that preserved key terms could still appear deceptively similar.

Unlike gold-explanation-anchored evaluations, *reference-free approaches* evaluate correctness directly against the explainer’s raw output. Ichmoukhamedov et al. [42] proposed Rank, Sign, and Value Agreement metrics to compare narrative claims against SHAP attributions. Pan et al. [69] evaluated the correctness of graph explanations via an information-theoretic measure, computing Point-wise Mutual Information between generated narratives and salient tokens from the model’s rationale. Since exact probability estimation is intractable, they approximate this by a masked language modeling setup, where explanations are judged by how well they aid recovery of masked essential tokens. Castellano et al. [14] aligned narrative text with Grad-CAM overlays and classification labels, employing CLIP-based [75] image–text similarity and S-BERT-based [77] text–label similarity. In a similar spirit, Yu et al. [108] assessed the correctness of their Knowledge-guided Forgery Detector method using Area Under the Curve (AUC) and Average Precision metrics on binary authenticity classification, complemented by GradCAM visualizations to examine whether the model’s attention corresponded to forged image regions. Similarly operating in the image domain, Wojciechowski et al. [99] assessed faithfulness through intervention-based tests, by masking image regions referenced in the narrative and observing the resulting changes in model confidence or predicted class. Performance degradation under these interventions provides direct, quantitative evidence of narrative–model alignment, offering an objective estimate of faithfulness.

Completeness. Human-centered evaluations have examined completeness by asking whether explanations capture the full range of relevant factors. Both Kočić et al. [51] and Zytek et al. [114] used 3-level anchored Likert scales to let participants rate how well narratives covered all pertinent information, an approach also adopted in the LLM-based evaluation in Zytek et al. [113] and Domnich et al. [25]. Automatic assessments, by contrast, infer completeness either from the coverage of elements found in expert-written gold narratives [12, 15, 42] or from alignment with explainer outputs. For instance, Castellano et al. [14] qualitatively judged completeness by checking whether narratives referred to all regions highlighted in Grad-CAM heatmaps. In contrast, Wojciechowski et al. [99] evaluated it via a stability analysis that compared explanations across input perturbations and class variations, using BLEU and METEOR scores. In this case, high intra-class similarity and low inter-class overlap indicate consistent coverage of relevant evidence and distinctiveness across categories, indirectly supporting narrative completeness. In their work, Shirvani-Mahdavi and Li [84] measure completeness through coverage metrics that quantify missed entities and missed relations, i.e., rule components present in the symbolic explainer but omitted from the generated narrative. These measures directly capture the extent to which the explanation covers all elements of the underlying rule. The authors also use automatic overlap metrics, such as ROUGE and BLEU, to further assess textual completeness relative to human-authored references.

From an empirical perspective, their findings confirm that fine-tuning and variable-type conditioning reduce omissions and improve coverage.

5.1.2 Linguistic Realization

Faithfulness alone does not ensure that an explanation is usable. The manner in which it is linguistically realized plays a central role. However, the quality of linguistic realization is seldom evaluated as an independent dimension. It is often assessed implicitly through human-centered measures such as comprehension, satisfaction, or trust, and thus frequently subsumed under what we term the Human Utility criteria. While acknowledging this partial overlap in practice, we distinguish the two dimensions conceptually: under LR, as described in this section, we include aspects primarily related to textual form and structure, and HU encompasses the user’s perceptual and affective response to the explanation.

Appropriateness of terminology. This criterion concerns whether the explanation employs language that is both understandable and contextually appropriate. Although often assessed implicitly within broader measures of clarity or comprehensibility, Giorgi et al. [30] explicitly instructed annotators to evaluate the suitability and readability of the terminology and phrasing used in the generated explanations using a Likert-scale assessment.

Fluency. This criterion refers to whether a narrative reads as natural and conversational. In human-centered evaluations, Zytek et al. [114] asked participants to rate the fluency of explanations, defining it as the degree to which a narrative sounded natural, while in their other work [113] they complement it with an LLM-based evaluation in which an evaluator model assigned rubric-based fluency scores. Wojciechowski et al. [99] likewise employed 5-point Likert ratings from human annotators to assess the fluency of generated narratives. In the absence of human evaluators or reference narratives, Shirvani-Mahdavi and Li [84] estimated fluency automatically using perplexity, which measures how well a language model predicts the next token in a text sequence. Lower perplexity values indicate smoother, more linguistically natural explanations.

Efficiency of expression. This criterion encompasses what the literature variously describes as conciseness or brevity, which partially overlaps with the more cognitive concept of complexity. It evaluates whether explanations achieve an effective balance between informativeness, length, and syntactic articulation. Automatic approaches are the most common in this category. Zytek et al. [114] operationalized efficiency through word count, interpreting shorter outputs as more concise. In a subsequent study, the same authors [113] introduced a continuous conciseness score derived from a deterministic word-length function normalized by the number of features. Pan et al. [69] proposed an Average Length Ratio to normalize brevity relative to the size of the input graph description. However, not all studies equate efficiency with shortness. Domnich et al. [25] explicitly manipulated narrative length to test whether perceived usefulness varied with complexity, finding that intermediate-length explanations were preferred. These findings indicate that expression efficiency is best understood as a balance between brevity and informativeness rather than as absolute shortness.

5.1.3 Human Utility

Finally, Human Utility groups criteria that capture the narrative’s effect on the user, both in terms of perception and practical value. This includes: (i) cognitive fit, covering plausibility, understandability, and internal consistency; (ii) attitudinal response, including trust, confidence, and satisfaction; and (iii) use value, encompassing actionability, insightfulness, and decision quality.

Cognitive fit. Human-centered evaluations have examined whether explanations are clear, plausible, and logically consistent. Giorgi et al. [30] assessed understandability, clarity, and plausibility using 5-point Likert-scale judgments,

a procedure also adopted by Wojciechowski et al. [99], who evaluated comprehensibility and plausibility, and by Shirvani-Mahdavi and Li [84], who applied the same scale to rate clarity. Kočić et al. [51] likewise targeted plausibility but used a 3-point anchored Likert scale with explicit level descriptions to ensure consistent interpretation among annotators. Cedro and Martens [16] included understandability within an eight-criterion evaluation framework, while Martens et al. [64] linked subjective clarity ratings to comprehension accuracy through task-based measures. Expanding this scope, Domnich et al. [25] incorporated understandability, feasibility, consistency, and complexity, demonstrating that both oversimplification and excessive detail can reduce explanatory utility. In a complementary direction, Zytek et al. [114] introduced context-awareness as an additional facet of cognitive fit, defined as the extent to which a narrative situates the explainer within a broader interpretive context. On the automatic side, Ichmoukhamedov et al. [42] proposed perplexity as a proxy for plausibility. On the automatic side, Ichmoukhamedov et al. [42] proposed using perplexity relative to a smaller LLM as a proxy for plausibility: they found that while this metric could identify unreasonable assumptions in an idealized setting, its results were far less conclusive for manipulated narratives. The authors suggest this is because LLMs are adept at making even manipulated assumptions sound plausible, indicating that perplexity alone is insufficient and that new automatic metrics are needed.

Attitudinal response. Another important dimension concerns the affective impact of explanations on users, particularly whether they foster trust, confidence, and satisfaction. Giorgi et al. [30] and Wojciechowski et al. [99] included overall satisfaction among their evaluation criteria, while Domnich et al. [25] explicitly added trust as a separate measure. Trust was also examined by Bhattacharya et al. [7], who found that perceptions of reliability varied based on participants’ prior knowledge. This indicates that attitudinal responses can be effectively captured through surveys but remain sensitive to users’ expectations and expertise. Cedro and Martens [16] assessed trustworthiness alongside confidence, satisfaction, and the perceived convincingness of narratives, whereas Martens et al. [64] extended this perspective by incorporating the likelihood of use and combining subjective ratings with behavioral indicators.

Use value. This criterion captures the practical contribution of explanations to user decision-making. Bhattacharya et al. [7] focused on both objective and subjective metrics to assess the usefulness of agent-augmented counterfactual explanations. *Perceived Actionability* [85] was assessed through three task-based evaluations—justification, how-to, and what-if tasks [55]—and complemented with 5-point Likert ratings to capture subjective perceptions. *Perceived Understandability* followed Hoffman et al. [40]’s framework, probing users’ confidence in interpreting and applying the explanations, while *Perceived Trust* was measured using Jian et al. [45]’s scale on automated systems. In parallel, *Perceived Taskload* was assessed using the NASA-TLX questionnaire [53], which accounts for cognitive, physical, and temporal demands. Cedro and Martens [16], instead, incorporated insightfulness and usability into their framework, linking narrative clarity to practical utility. Martens et al. [64] extended this line by measuring decision quality and response time to explicitly connect explanatory quality with task performance. In the reinforcement learning setting, Xi-Jia et al. [100] examined whether explanations enhance users’ ability to predict an agent’s following action. Participants performed prediction tasks both with and without explanations and provided pairwise preference ratings on a 5-point Likert scale. Results indicated significantly higher prediction accuracy and stronger preferences for LLM-generated narratives grounded in behavioral representations, which achieved parity with human-written explanations.

5.2 Emerging Direction: LLM as Human-like Evaluators

A recent methodological development concerns the use of LLMs not only as judges of explanations but also as surrogates for human participants in evaluation tasks. These studies explore whether LLMs can approximate or even substitute

human judgments in assessing explanations, thereby reducing the cost and effort associated with large-scale user studies. This line of research marks a shift from evaluating the explanations themselves to evaluating the evaluators, that is, examining to what extent LLM-based assessments are compatible with, or predictive of, human preferences. Insights from these works also inform studies employing LLM-as-Judge frameworks, where the alignment between automated and human ratings becomes a central indicator of reliability and validity in explanation evaluation. We recall the work by Shirvani-Mahdavi and Li [84] and Domnich et al. [25], where LLMs were employed to approximate human judgments in explanation evaluation tasks. While Shirvani-Mahdavi and Li [84] explored alignment between model- and human-based evaluators in rule-to-text explanations, Domnich et al. [25] extended this approach to counterfactual explanations, demonstrating strong correspondence between fine-tuned LLM assessments and aggregated human ratings across multiple explanatory dimensions.

De Bona et al. [22] take this a step further by replicating an existing user study [17] with LLMs replacing human participants, directly testing whether models can serve as stand-ins for human subjects in explanation evaluation tasks. Their study compares causal and counterfactual explanations under high vs. low familiarity conditions across functions such as helpfulness ratings, prediction, and confidence. The authors introduce two notions of alignment: general alignment, i.e., whether the statistical conclusions of the human study are replicated, and absolute alignment, i.e., whether the mean ratings themselves match. They find that LLMs reproduce most aggregate patterns, particularly on prediction tasks, but alignment breaks down on confidence-related judgments. Interestingly, model size did not guarantee better performance: smaller models sometimes aligned more closely with humans than larger ones. Moreover, methodological choices regarding memory usage and prompting conditions substantially influenced alignment outcomes, underscoring the approach’s potential instability.

Domnich et al. [25] instead explicitly fine-tuned LLMs to predict human ratings of counterfactual explanations. They collect judgments from 206 participants across 30 scenarios and eight evaluation dimensions (feasibility, coherence, trust, complexity, fairness, completeness, understandability, and satisfaction), creating the CounterEval dataset. Fine-tuned LLMs achieve up to 85% accuracy in predicting average ratings, with strong performance on completeness, complexity, and satisfaction, though feasibility and trust remained difficult to capture. Beyond aggregate scores, the authors show that LLMs can approximate individual evaluative patterns. For three out of four participants, fine-tuned models reproduced personal rating tendencies with about 90% accuracy, demonstrating the potential to model not only collective but also individual human judgments. However, beyond the limitation of the small dataset acknowledged by the authors, it should be noted that collapsing the 6-point Likert ratings into three classes simplified modeling but risked obscuring the finer-grained distinctions in human judgments.

Overall, although still emerging in XAI, this line of research raises critical methodological questions for the future of XAI evaluation. Indeed, if LLMs can reliably approximate human judgments, they may offer a scalable alternative to costly user studies, enabling rapid iteration and broader experimental coverage. Nevertheless, concerns remain about domain transfer, susceptibility to training data leakage, and biases introduced by prompt design or model architecture. The authors themselves highlight these limitations and call for more systematic investigation before LLMs can be considered trustworthy replacements for human evaluators.

5.3 Pros and Cons of XAI Narrative Evaluation Methods

Automatic metrics offer scalability and objectivity, making them attractive for evaluating narrative explanations. However, their use also presents essential caveats. Reference-free metrics, such as perplexity, are highly sensitive to confounding factors, such as spelling variations or uncommon formulations, that are unrelated to explanatory quality.

As stressed by [42], narratives with unexpectedly high perplexity values should therefore be subject to closer inspection, but perplexity alone cannot serve as an exclusive measure of explanation quality. While such metrics avoid dependence on curated gold standards, they risk rewarding superficial fluency or penalizing stylistic diversity, and thus must be interpreted with care.

By contrast, reference-based metrics anchor evaluation to a set of gold-standard explanations. Lexical measures such as BLEU, ROUGE, or METEOR assess surface-level overlap, while semantic embedding-based metrics such as BLEURT or cosine similarity attempt to capture meaning alignment more flexibly. Their main strength lies in standardization and replicability: when a high-quality reference set exists, they allow systematic comparison across models and datasets. However, they also inherit biases from the reference corpus itself. If expert-written references reflect specific stylistic preferences, cultural assumptions, or domain framings, models that deviate from these—even while producing faithful or more user-friendly explanations—may be unfairly penalized. In this sense, reference-based metrics risk conflating stylistic conformity with explanatory quality and may struggle in domains where multiple plausible narratives exist. Moreover, as stressed by Ichmoukamedov et al. [42], embedding-based metrics, such as BLEURT, provide similarity scores but little diagnostic insight: they may signal divergence without clarifying whether it stems from harmless stylistic variation or from deeper semantic errors. Finally, even when automatic metrics explicitly target faithfulness, their operationalization is non-trivial: numerical attributions must be translated into natural language, and design choices – such as whether to include exact SHAP values or only the rank ordering of features – directly affect what the “alignment” with the explainer entails. These challenges highlight that while automatic metrics are valuable, they should ideally be complemented by human-centered or hybrid evaluation strategies that capture aspects of explanatory quality beyond surface-level similarity or statistical likelihood. This is further complicated by the fact that, even if technically correct, an explanation is necessary but not sufficient for an adequate explanation from a real-world user’s perspective. This is precisely where user studies become essential, as they provide a complementary perspective for assessing the experiential and pragmatic aspects of narrative explanations. They are indispensable for evaluating dimensions such as comprehensibility, trust, perceived usefulness, and actual actionability, which are qualities that cannot be reliably inferred from textual similarity or model alignment alone. However, the interpretive power of such studies is undermined by both methodological and structural weaknesses. A systematic review by Pičulin et al. [73] reveals that many studies lack reproducibility, fail to use proper baselines, and rely on low-fidelity “toy” tasks that do not reflect real-world decision-making. An overreliance further compromises generalizability on non-representative participant pools, such as crowdsourced workers or students, rather than the domain experts who would actually use the system.

Furthermore, subjective ratings themselves can be highly misleading. Users may rate an explanation as “clear” simply because it aligns with their expectations, even if it misrepresents the model’s actual reasoning. This relates to the XAI “placebo effect,” where the mere presence of an explanation—even an irrelevant one—can inflate user trust. Critically, Pičulin et al. [73] highlight a documented disconnect between these subjective perceptions and objective outcomes: self-reported trust and understanding are poor predictors of actual task performance. This asymmetry implies that explanations that merely sound plausible may be rated favorably even when factually inaccurate, a phenomenon related to the “plausibility hallucination” effect described by Ichmoukamedov et al. [42]. This is further compounded by the inherent difficulty in separating subjective preferences from user evaluations of verifiable aspects. For all these reasons, while human evaluations are irreplaceable, they must be designed with far greater rigor to disentangle perceptual clarity from factual correctness and to provide reliable evidence of the actual value of explanations.

LLM-based assessments occupy an intermediate position between these two poles, combining the scalability of automatic metrics with a degree of qualitative interpretability. However, these benefits are critically dependent on

calibration and validation. Moreover, as discussed in Section 5.2, alignment between LLM and human ratings is often strong in relative rankings but weaker in absolute scores, fine-grained assessments, and experiential aspects such as trust. Furthermore, LLM-based evaluators are susceptible to framing effects and prompt sensitivity. For this reason, hybrid frameworks where LLM judges are validated against human annotators and complemented by objective metrics are emerging as a promising direction for achieving both interpretive validity and methodological efficiency. As emphasized by Domnich et al. [25], such hybrid paradigms should involve iterative retraining with newer LLMs and retain human oversight “in the loop” to preserve normative grounding and avoid automation bias in evaluative feedback.

6 Prompting Strategies

Prompt design is a decisive factor in shaping LLM behavior. Because models interpret textual instructions directly, even subtle lexical or syntactic changes can dramatically affect output quality, reasoning structure, or factual accuracy. This sensitivity exposes a core challenge: the mapping between prompt and model behavior remains opaque and unstable. As a result, *prompt engineering* has become a critical methodological field focused on designing, optimizing, and evaluating prompts that elicit faithful and coherent responses. In XAI Narratives, where interpretive clarity is as vital as performance, practical prompt engineering ensures that generated explanations remain both faithful to model reasoning and intelligible to human users.

6.1 Current Prompting Trends

Prompting strategies shape how users (or systems) interact with LLMs and, in XAI Narratives, largely determine explanatory style, faithfulness, and depth. The field has moved from simple instructions to multi-stage deliberation. We describe the families most relevant to XAI and summarize them, along with illustrative examples, in Table 3.

Role Prompting. Role prompting [52] sets the model’s identity and stance to control tone, granularity, and reasoning depth (e.g., “You are an explainable AI assistant specialized in interpreting model decisions. Explain the following prediction.”). This helps align explanations with domain norms and pedagogical clarity.

Zero-Shot Prompting. Zero-shot prompting supplies only an instruction; the model relies on pretraining to infer the task (e.g., “Explain why the model predicted label A instead of B, citing the most relevant features.”). It is useful when examples are unavailable, though reasoning may be shallow on domain-specific tasks.

Few-Shot Prompting. Few-shot prompting embeds a handful of exemplars to fix format, vocabulary, and reasoning style (e.g., “Example 1 ... Example 2 ... Now explain the following case using the same style and level of detail.”). In XAI, this improves consistency of explanatory structure (lists, JSON) and domain semantics.

Chain-of-Thought (CoT). CoT prompting elicits intermediate reasoning before the answer [98] (e.g., “Explain the model’s prediction step by step, showing the intermediate reasoning before the final explanation.”). This supports transparency and causal interpretability by making the inference path explicit.

Tree-of-Thought (ToT). ToT generalizes CoT to branching deliberation [104], exploring alternatives and selecting the most coherent (e.g., “Explore different possible explanations ... compare the reasoning paths and choose the most coherent one.”). This is valuable for multi-hop or graph-based XAI, where multiple narratives can justify an outcome.

7 Conclusion and Future Directions

This work proposed a formal and conceptual framework for *XAI Narratives*, defining the Narrative Generation Problem (NGP) as the task of transforming technical explanations into linguistically coherent, contextually grounded accounts.

Table 3. Prompting strategies for model explanation, including their purposes and illustrative prompts.

Strategy	Purpose	Illustrative Prompt
Role Prompting	Defines the model’s identity, aligning tone and granularity with the explanation goal.	<i>You are an explainable AI assistant specialized in interpreting model decisions. Explain the following prediction.</i>
Zero-Shot	Provides only a task instruction and relies on the model’s intrinsic knowledge.	<i>Explain why the model predicted label A instead of B, citing the most relevant features.</i>
Few-Shot	Gives a few examples of task execution to guide style and reasoning.	<i>Example 1 – Input: X, Output: Y, Explanation: Z Example 2 – Input: X, Output: Y, Explanation: Z Now, explain the following case using the same style and level of detail. Input: X, Output: Y.</i>
Chain-of-Thought	Instructs the model to do intermediate reasoning steps before answering.	<i>Explain the model’s prediction step by step, showing the intermediate reasoning before giving the final explanation.</i>
Tree-of-Thought	Explores multiple reasoning paths and selects the most coherent one.	<i>Explore different possible explanations for the prediction on this target node. Compare the reasoning paths and choose the most coherent one.</i>

Through a structured analysis of the literature, we examined how Large Language Models (LLMs) have been employed to instantiate this process across domains, identifying both methodological advances and persistent challenges.

Our survey reveals that while LLMs inaugurate a new class of explanation interfaces—capable of producing fluent, interactive, and context-sensitive narratives—the reliability and interpretive soundness of these outputs remain contingent on dimensions that extend far beyond linguistic fluency or syntactic correctness. Moreover, the interpretive quality of such narratives is influenced by factors such as prompt design, grounding fidelity, the integration of XAI modules, and the model’s capacity to adapt explanations to users’ cognitive and domain-specific contexts. In light of these observations, the following research trajectories appear particularly promising.

Refining Narrative structure beyond descriptive accounts. Current XAI Narratives often mirror the correlational logic of the underlying explainer, resulting in outputs that enumerate associations rather than articulate relations among them. Future research should focus on refining narrative formation to move beyond description toward richer structural representations that encode causal, temporal, and conditional dependencies among explanatory elements. This does not entail endowing LLMs with the ability to infer causal truths about the world, but rather ensuring that the generated narratives respect and clarify the relational structure already implicit in the explainer’s output. By aligning narrative coherence with epistemic fidelity, XAI Narratives can evolve from surface-level reformulations into communicatively structured accounts that more faithfully convey the logic and dynamics of model behavior.

Toward principled and transparent prompting. The quality and reliability of XAI Narratives are profoundly shaped by prompt design. However, prompting practices remain ad hoc and often irreproducible. Establishing methodological standards for prompt documentation, sharing, and benchmarking is essential to transform prompt engineering from an artisanal craft into a reproducible scientific practice. Future research should also explore adaptive, self-correcting prompting strategies that can dynamically learn domain and user context.

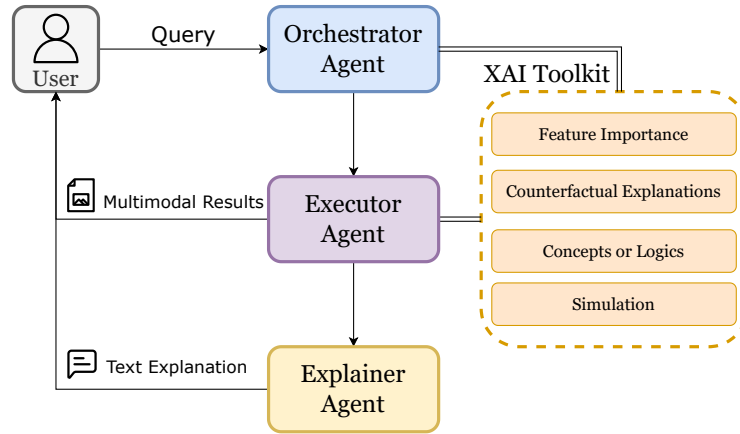


Fig. 4. *Generalized Multi-Agent architecture for XAI Narratives.* The framework illustrates an agentic pipeline for generating narrative explanations across domains. The Orchestrator Agent mediates between the user’s query, the XAI Toolkit (a library of interpretability primitives), and the Executor Agent, which operationalizes these methods. The Explainer Agent transforms the resulting interpretive artefacts into coherent textual narratives.

Advancing evaluation paradigms. Evaluation of narrative explanations remains fragmented across epistemic, linguistic, and user-centered dimensions. A key direction lies in developing integrated frameworks that balance faithfulness to model reasoning, linguistic quality, and pragmatic utility. Hybrid evaluation pipelines—combining human judgments, LLM-based scoring, and behavioral studies—can help ensure that future metrics remain grounded in human understanding and trust rather than in LLM self-assessment only.

Toward agentic and context-aware explanation pipelines. Finally, as AI systems grow increasingly agentic and interactive, XAI Narratives are poised to evolve from static, post-hoc outputs into adaptive communicative acts within multi-agent ecosystems. Future research should explore how agentic AI pipelines can serve as universal explanatory frameworks, integrating distinct yet coordinated components: an *orchestrator agent* that interprets user intent; an *XAI Toolkit* comprising diverse explanation methods; an *executor agent* that interfaces with underlying models; and an *explainer agent* that generates coherent narrative accounts. Such an architecture (see Figure 4) would enable dynamic selection of the most suitable XAI technique in response to a user’s query, supporting closed-loop reasoning, interactive evaluation, and socially grounded justification of AI behavior.

In summary, advancing XAI Narratives requires moving beyond surface-level translation toward epistemically grounded, contextually adaptive, and methodologically transparent systems. By bridging causal reasoning, principled prompting, rigorous evaluation, and agentic interaction, future research can transform LLM-based narratives from persuasive descriptions into genuinely interpretable explanations—aligning the communicative capacity of language models with the foundational goals of trustworthy and socially responsive AI.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).

- [2] Sajid Ali, Tamer Abuhmed, Shaker El-Sappagh, Khan Muhammad, Jose M Alonso-Moral, Roberto Confalonieri, Riccardo Guidotti, Javier Del Ser, Natalia Diaz-Rodriguez, and Francisco Herrera. 2023. Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence. *Information fusion* 99 (2023), 101805.
- [3] Sebastian Bach, Alexander Binder, Grégoire Montavon, Frederick Klauschen, Klaus-Robert Müller, and Wojciech Samek. 2015. On Pixel-Wise Explanations for Non-Linear Classifier Decisions by Layer-Wise Relevance Propagation. *PLoS ONE* 10 (2015). <https://api.semanticscholar.org/CorpusID:9327892>
- [4] Satanejeev Banerjee and Alon Lavie. 2005. METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In *Proceedings of the acl workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization*. 65–72.
- [5] Barry Becker and Ronny Kohavi. 1996. Adult. UCI Machine Learning Repository. DOI: <https://doi.org/10.24432/C5XW20>.
- [6] Leonardo Berti, Flavio Giorgi, and Gjergji Kasneci. 2025. Emergent abilities in large language models: A survey. *arXiv preprint arXiv:2503.05788* (2025).
- [7] Aditya Bhattacharya, Tim Vanherwegen, and Katrien Verbert. 2025. "Show Me How": Benefits and Challenges of Agent-Augmented Counterfactual Explanations for Non-Expert Users. In *Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization*. 174–184.
- [8] Ahsan Bilal, David Ebert, and Beiyu Lin. 2025. Llm for explainable ai: A comprehensive survey. *arXiv preprint arXiv:2504.00125* (2025).
- [9] Alexander Binder, Grégoire Montavon, Sebastian Bach, Klaus-Robert Müller, and Wojciech Samek. 2016. Layer-wise Relevance Propagation for Neural Networks with Local Renormalization Layers. *arXiv:1604.00825 [cs.CV]* <https://arxiv.org/abs/1604.00825>
- [10] Antoine Bordes, Nicolas Usunier, Alberto Garcia-Duran, Jason Weston, and Oksana Yakhnenko. 2013. Translating Embeddings for Modeling Multi-relational Data. In *Advances in Neural Information Processing Systems*, C.J. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K.Q. Weinberger (Eds.), Vol. 26. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2013/file/1cecc7a77928ca8133fa24680a88d2f9-Paper.pdf
- [11] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.
- [12] James Burton, Noura Al Moubayed, and Amir Enshaee. 2023. Natural language explanations for machine learning classification decisions. In *2023 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 1–9.
- [13] Erik Cambria, Lorenzo Malandri, Fabio Mercorio, Mario Mezzananza, and Navid Nobani. 2023. A survey on XAI and natural language explanations. *Information Processing & Management* 60, 1 (2023), 103111.
- [14] Giovanna Castellano, Maria Grazia Miccoli, Raffaele Scaringi, Gennaro Vessio, Gianluca Zaza, et al. 2024. Using LLMs to explain AI-generated art classification via Grad-CAM heatmaps. In *Proceedings of 5th Italian Workshop on Explainable Artificial Intelligence, co-located with the 23rd International Conference of the Italian Association for Artificial Intelligence, Bolzano, Italy*.
- [15] Alessandro Castelnovo, Roberto Depalmas, Fabio Mercorio, Nicolò Mombelli, Daniele Poterti, Antonio Serino, Andrea Seveso, Salvatore Sorrentino, and Laura Viola. 2024. Augmenting XAI with LLMs: A Case Study in Banking Marketing Recommendation. In *Explainable Artificial Intelligence*, Luca Longo, Sebastian Lapuschkin, and Christin Seifert (Eds.). Springer Nature Switzerland, Cham, 211–229.
- [16] Mateusz Cedro and David Martens. 2024. GraphXAIN: Narratives to Explain Graph Neural Networks. *arXiv preprint arXiv:2411.02540* (2024).
- [17] Lenart Celar and Ruth MJ Byrne. 2023. How people reason with counterfactual and causal explanations for artificial intelligence decisions in familiar and unfamiliar domains. *Memory & Cognition* 51, 7 (2023), 1481–1496.
- [18] Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E. Gonzalez, Ion Stoica, and Eric P. Xing. 2023. Vicuna: An Open-Source Chatbot Impressing GPT-4 with 90%* ChatGPT Quality. <https://lmsys.org/blog/2023-03-30-vicuna/>
- [19] Fabio Chiusi, Brigitte Alfter, Minna Ruckenstein, and Tuukka Lehtiniemi. 2020. Automating society report 2020. (2020).
- [20] Lea Cohausz. 2022. Towards Real Interpretability of Student Success Prediction Combining Methods of XAI and Social Science. *International Educational Data Mining Society* (2022).
- [21] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. 2023. InstructBLIP: Towards General-purpose Vision-Language Models with Instruction Tuning. *arXiv:2305.06500 [cs.CV]* <https://arxiv.org/abs/2305.06500>
- [22] Francesco Bombassei De Bona, Gabriele Dominici, Tim Miller, Marc Langheinrich, and Martin Gjoreski. 2024. Evaluating explanations through llms: Beyond traditional user studies. *arXiv preprint arXiv:2410.17781* (2024).
- [23] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *arXiv:1810.04805 [cs.CL]* <https://arxiv.org/abs/1810.04805>
- [24] Amit Dhurandhar, Pin-Yu Chen, Ronny Luss, Chun-Chen Tu, Paishun Ting, Karthikeyan Shanmugam, and Payel Das. 2018. Explanations based on the Missing: Towards Contrastive Explanations with Pertinent Negatives. *arXiv:1802.07623 [cs.AI]* <https://arxiv.org/abs/1802.07623>
- [25] Marharyta Domnich, Julius Völja, Rasmus Moorits Veski, Giacomo Magnifico, Kadi Tulver, Eduard Barbu, and Raul Vicente. 2025. Towards unifying evaluation of counterfactual explanations: Leveraging large language models for human-centric assessments. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 16308–16316.
- [26] Thomas Cason Frank E. Harrell Jr. 2017. *Titanic dataset*. <https://www.openml.org/d/40945>
- [27] Arturo Fredes and Jordi Vitria. 2024. Using LLMs for Explaining Sets of Counterfactual Examples to Final Users. *arXiv preprint arXiv:2408.15133* (2024).

- [28] Luis Antonio Galárraga, Christina Teflioudi, Katja Hose, and Fabian Suchanek. 2013. AMIE: association rule mining under incomplete evidence in ontological knowledge bases. In *Proceedings of the 22nd International Conference on World Wide Web (Rio de Janeiro, Brazil) (WWW '13)*. Association for Computing Machinery, New York, NY, USA, 413–422. <https://doi.org/10.1145/2488388.2488425>
- [29] Luis Antonio Galárraga, Christina Teflioudi, Katja Hose, and Fabian Suchanek. 2013. AMIE: association rule mining under incomplete evidence in ontological knowledge bases. In *Proceedings of the 22nd international conference on World Wide Web*. 413–422.
- [30] Flavio Giorgi, Cesare Campagnano, Fabrizio Silvestri, and Gabriele Tolomei. 2025. Natural Language Counterfactual Explanations for Graphs Using Large Language Models. In *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics (Proceedings of Machine Learning Research, Vol. 258)*, Yingzhen Li, Stephan Mandt, Shipra Agrawal, and Emtiyaz Khan (Eds.). PMLR, 3565–3573. <https://proceedings.mlr.press/v258/giorgi25a.html>
- [31] Flavio Giorgi, Matteo Silvestri, Cesare Campagnano, Fabrizio Silvestri, and Gabriele Tolomei. 2025. Enhancing XAI Narratives through Multi-Narrative Refinement and Knowledge Distillation. *arXiv preprint arXiv:2510.03134* (2025).
- [32] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. 2024. The Llama 3 Herd of Models. *arXiv:2407.21783 [cs.AI]* <https://arxiv.org/abs/2407.21783>
- [33] Riccardo Guidotti. 2024. Counterfactual explanations and how to find them: literature review and benchmarking. *Data Mining and Knowledge Discovery* 38, 5 (2024), 2770–2824.
- [34] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *arXiv:2501.12948 [cs.CL]* <https://arxiv.org/abs/2501.12948>
- [35] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. 2018. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. *arXiv:1801.01290 [cs.LG]* <https://arxiv.org/abs/1801.01290>
- [36] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 770–778. <https://doi.org/10.1109/CVPR.2016.90>
- [37] Yinhan He, Zaiyi Zheng, Patrick Soga, Yaozhen Zhu, Jundong Li, et al. 2024. Explaining Graph Neural Networks with Large Language Models: A Counterfactual Perspective for Molecular Property Prediction. *arXiv preprint arXiv:2410.15165* (2024).
- [38] Evan Hernandez, Sarah Schwettmann, David Bau, Teona Bagashvili, Antonio Torralba, and Jacob Andreas. 2022. Natural Language Descriptions of Deep Visual Features. *arXiv:2201.11114 [cs.CV]* <https://arxiv.org/abs/2201.11114>
- [39] James Hinns and David Martens. 2024. Exposing image classifier shortcuts with counterfactual frequency (cof) tables. *arXiv preprint arXiv:2405.15661* (2024).
- [40] Robert R Hoffman, Shane T Mueller, Gary Klein, and Jordan Litman. 2018. Metrics for explainable AI: Challenges and prospects. *arXiv preprint arXiv:1812.04608* (2018).
- [41] Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. 2024. Gpt-4o system card. *arXiv preprint arXiv:2410.21276* (2024).
- [42] Timour Ichmourkamedov, James Hinns, and David Martens. 2024. How good is my story? Towards quantitative metrics for evaluating LLM-generated XAI narratives. *arXiv preprint arXiv:2412.10220* (2024).
- [43] Fred Jelinek, Robert L Mercer, Lalit R Bahl, and James K Baker. 1977. Perplexity—a measure of the difficulty of speech recognition tasks. *The Journal of the Acoustical Society of America* 62, S1 (1977), S63–S63.
- [44] Jongheon Jeong, Yang Zou, Taewan Kim, Dongqing Zhang, Avinash Ravichandran, and Onkar Dabeer. 2023. Winclip: Zero-/few-shot anomaly classification and segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 19606–19616.
- [45] Jiun-Yin Jian, Ann M Bisantz, and Colin G Drury. 2000. Foundations for an empirically determined scale of trust in automated systems. *International journal of cognitive ergonomics* 4, 1 (2000), 53–71.
- [46] AQ Jiang, A Sablayrolles, A Mensch, C Bamford, DS Chaplot, D de las Casas, F Bressand, G Lengyel, G Lample, L Saulnier, et al. 2023. Mistral 7B (2023). *arXiv preprint arXiv:2310.06825* (2023).
- [47] Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, Léo Renard Lavaud, Lucile Saulnier, Marie-Anne Lachaux, Pierre Stock, Sandeep Subramanian, Sophia Yang, Szymon Antoniak, Teven Le Scao, Théophile Gervet, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2024. Mixtral of Experts. *arXiv:2401.04088 [cs.LG]* <https://arxiv.org/abs/2401.04088>
- [48] Vidhya Kamakshi and Narayanan C. Krishnan. 2023. Explainable Image Classification: The Journey So Far and the Road Ahead. *AI* 4, 3 (2023), 620–651. <https://doi.org/10.3390/ai4030033>
- [49] Abderrazak Khediri, Hamda Slimi, Ayoub Yahiaoui, Makhlof Derdour, Hakim Bendjenna, and Charaf Eddine Ghenai. 2024. Enhancing machine learning model interpretability in intrusion detection systems through SHAP explanations and LLM-generated descriptions. In *2024 6th International Conference on Pattern Analysis and Intelligent Systems (PAIS)*. IEEE, 1–6.
- [50] Haechang Kim, Hao Chen, Can Li, and Jong Min Lee. 2025. TalkToAgent: A Human-centric Explanation of Reinforcement Learning Agents with Large Language Models. *arXiv:2509.04809 [cs.AI]* <https://arxiv.org/abs/2509.04809>
- [51] Veljka Kočić, Niko Lukač, Dž Rožajac, Stefan Schweng, Christoph Gollub, Arne Nothdurft, Karl Stampfer, Javier Del Ser, and Andreas Holzinger. 2025. LLM in the Loop: A Framework for Contextualizing Counterfactual Segment Perturbations in Point Clouds. *IEEE access* (2025).
- [52] Aobo Kong, Shiwang Zhao, Hao Chen, Qicheng Li, Yong Qin, Ruiqi Sun, Xin Zhou, Enzhi Wang, and Xiaohang Dong. 2024. Better Zero-Shot Reasoning with Role-Play Prompting. *arXiv:2308.07702 [cs.CL]* <https://arxiv.org/abs/2308.07702>

- [53] Todd Kulesza, Simone Stumpf, Margaret Burnett, Weng-Keen Wong, Yann Riche, Travis Moore, Ian Oberst, Amber Shinsell, and Kevin McIntosh. 2010. Explanatory debugging: Supporting end-user debugging of machine-learned programs. In *2010 IEEE Symposium on Visual Languages and Human-Centric Computing*. IEEE, 41–48.
- [54] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. 2019. BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. *arXiv:1910.13461 [cs.CL]* <https://arxiv.org/abs/1910.13461>
- [55] Brian Y Lim, Anind K Dey, and Daniel Avrahami. 2009. Why and why not explanations improve the intelligibility of context-aware intelligent systems. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 2119–2128.
- [56] Chin-Yew Lin. 2004. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*. 74–81.
- [57] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. 2014. Microsoft coco: Common objects in context. In *European conference on computer vision*. Springer, 740–755.
- [58] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. 2024. LLaVA-NeXT: Improved reasoning, OCR, and world knowledge. <https://llava-vl.github.io/blog/2024-01-30-llava-next/>
- [59] Pedro Lopes, Eduardo Silva, Cristiana Braga, Tiago Oliveira, and Luis Rosado. 2022. XAI systems evaluation: A review of human and computer-centred methods. *Applied Sciences* 12, 19 (2022), 9423.
- [60] Ana Lucic, Maartje A Ter Hoeve, Gabriele Tolomei, Maarten De Rijke, and Fabrizio Silvestri. 2022. Cf-gnnexplainer: Counterfactual explanations for graph neural networks. In *International Conference on Artificial Intelligence and Statistics*. PMLR, 4499–4511.
- [61] Scott M Lundberg and Su-In Lee. 2017. A unified approach to interpreting model predictions. *Advances in neural information processing systems* 30 (2017).
- [62] Haiping Ma, Yajing Zhang, Shengyi Sun, Ting Liu, and Yu Shan. 2023. A comprehensive survey on NSGA-II for multi-objective optimization and applications. *Artificial Intelligence Review* 56, 12 (2023), 15217–15270.
- [63] Jing Ma, Ruocheng Guo, Saumitra Mishra, Aidong Zhang, and Jundong Li. 2022. Clear: Generative counterfactual explanations on graphs. *Advances in neural information processing systems* 35 (2022), 25895–25907.
- [64] David Martens, James Hinns, Camille Dams, Mark Vergouwens, and Theodoros Evgeniou. 2025. Tell me a story! Narrative-driven XAI with Large Language Models. *Decision Support Systems* 191 (2025), 114402.
- [65] Harry Mayne, Ryan Othniel Kearns, Yushi Yang, Andrew M Bean, Eoin Delaney, Chris Russell, and Adam Mahdi. 2025. LLMs Don't Know Their Own Decision Boundaries: The Unreliability of Self-Generated Counterfactual Explanations. *arXiv preprint arXiv:2509.09396* (2025).
- [66] Prasant Kumar Mohanty, Sharmila Anand John Francis, Rabindra Kumar Barik, K Hemant Kumar Reddy, Diptendu Sinha Roy, and Manob Jyoti Saikia. 2025. IMPACT: an interactive multi-disease prevention and counterfactual treatment system using explainable AI and a multimodal LLM. *PeerJ Computer Science* 11 (2025), e2839.
- [67] Ramaravind K Mothilal, Amit Sharma, and Chenhao Tan. 2020. Explaining machine learning classifiers through diverse counterfactual explanations. In *Proceedings of the 2020 conference on fairness, accountability, and transparency*. 607–617.
- [68] Matthew J Page, Joanne E McKenzie, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, et al. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *International journal of surgery* 88 (2021), 105906.
- [69] Bo Pan, Zhen Xiong, Guanchen Wu, Zheng Zhang, Yifei Zhang, and Liang Zhao. 2024. GraphNarrator: Generating Textual Explanations for Graph Neural Networks. *arXiv preprint arXiv:2410.15268* (2024).
- [70] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*. 311–318.
- [71] European Parliament and Council of the European Union. 2024. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonized rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828. Official Journal of the European Union, L 1689, 12 July 2024, pp. 1–241.
- [72] Zhiliang Peng, Wenhui Wang, Li Dong, Yaru Hao, Shaohan Huang, Shuming Ma, and Furu Wei. 2023. Kosmos-2: Grounding Multimodal Large Language Models to the World. *arXiv:2306.14824 [cs.CL]* <https://arxiv.org/abs/2306.14824>
- [73] Matej Pičulin, Bernarda Petek, Irena Ograjensek, and Erik Strumbelj. [n. d.]. Position: Explainable AI Cannot Advance Without Better User Studies. In *Forty-second International Conference on Machine Learning Position Paper Track*.
- [74] Qwen Team Qwen Team. 2024. Qwen2.5: A Party of Foundation Models. <https://qwenlm.github.io/blog/qwen2.5/>
- [75] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PmlR, 8748–8763.
- [76] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2023. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *arXiv:1910.10683 [cs.LG]* <https://arxiv.org/abs/1910.10683>
- [77] Nils Reimers and Iryna Gurevych. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084* (2019).
- [78] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2016. "Why should i trust you?" Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. 1135–1144.

- [79] Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. 2011. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, 627–635.
- [80] Fawaz Sammani, Tanmoy Mukherjee, and Nikos Deligiannis. 2022. NLX-GPT: A Model for Natural Language Explanations in Vision and Vision-Language Tasks. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 8312–8322. <https://doi.org/10.1109/CVPR52688.2022.00814>
- [81] Gesina Schwalbe and Bettina Finzel. 2024. A comprehensive taxonomy for explainable artificial intelligence: a systematic survey of surveys on methods and concepts. *Data Mining and Knowledge Discovery* 38, 5 (2024), 3043–3101.
- [82] Thibault Sellam, Dipanjan Das, and Ankur P Parikh. 2020. BLEURT: Learning robust metrics for text generation. *arXiv preprint arXiv:2004.04696* (2020).
- [83] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. 2020. Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *International Journal of Computer Vision* 128, 2 (2020), 336–359. <https://doi.org/10.1007/s11263-019-01228-7>
- [84] Nasim Shirvani-Mahdavi and Chengkai Li. 2025. Rule2Text: A Framework for Generating and Evaluating Natural Language Explanations of Knowledge Graph Rules. *arXiv:2508.10971 [cs.CL]* <https://arxiv.org/abs/2508.10971>
- [85] Sarah J Shoemaker, Michael S Wolf, and Cindy Brach. 2014. Development of the Patient Education Materials Assessment Tool (PEMAT): a new measure of understandability and actionability for print and audiovisual patient information. *Patient education and counseling* 96, 3 (2014), 395–403.
- [86] Matteo Silvestri, Flavio Giorgi, Fabrizio Silvestri, and Gabriele Tolomei. 2025. Evaluating Latent Knowledge of Public Tabular Datasets in Large Language Models. *arXiv:2510.20351 [cs.CL]* <https://arxiv.org/abs/2510.20351>
- [87] Timo Speith. 2022. A review of taxonomies of explainable artificial intelligence (XAI) methods. In *Proceedings of the 2022 ACM conference on fairness, accountability, and transparency*. 2239–2250.
- [88] Timo Speith and Markus Langer. 2023. A new perspective on evaluation methods for explainable artificial intelligence (xai). In *2023 IEEE 31st International Requirements Engineering Conference Workshops (REW)*. IEEE, 325–331.
- [89] Ashley Suh, Kenneth Alperin, Harry Li, and Steven R Gomez. 2025. Don’t Just Translate, Agitate: Using Large Language Models as Devil’s Advocates for AI Explanations. In *Proceedings of the 5th HCXAI Workshop*. Zenodo, 48–55. <https://doi.org/10.5281/ZENODO.15170455>
- [90] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. 2017. Axiomatic Attribution for Deep Networks. *arXiv:1703.01365 [cs.LG]* <https://arxiv.org/abs/1703.01365>
- [91] Vinita Swamy, Davide Romano, Bhargav Srinivasa Desikan, Oana-Maria Camburu, and Tanja Käser. 2025. iLLuMinaTE: An LLM-XAI Framework Leveraging Social Science Explanation Theories Towards Actionable Student Performance Feedback. *arXiv:2409.08027 [cs.CY]* <https://arxiv.org/abs/2409.08027>
- [92] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. 2023. Gemini: A Family of Highly Capable Multimodal Models. *arXiv:2312.11805 [cs.CL]* <https://arxiv.org/abs/2312.11805>
- [93] Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al. 2024. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv:2403.05530 [cs.CL]* <https://arxiv.org/abs/2403.05530>
- [94] Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, et al. 2024. Gemma 2: Improving Open Language Models at a Practical Size. *arXiv:2408.00118 [cs.CL]* <https://arxiv.org/abs/2408.00118>
- [95] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open Foundation and Fine-Tuned Chat Models. *arXiv:2307.09288 [cs.CL]* <https://arxiv.org/abs/2307.09288>
- [96] Paul Voigt and Axel Von dem Bussche. 2017. The eu general data protection regulation (gdpr). *A Practical Guide, 1st Ed., Cham: Springer International Publishing* 10, 3152676 (2017), 10–5555.
- [97] Sandra Wachter, Brent Mittelstadt, and Chris Russell. 2018. Counterfactual Explanations without Opening the Black Box: Automated Decisions and the GDPR. *arXiv:1711.00399 [cs.AI]* <https://arxiv.org/abs/1711.00399>
- [98] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2023. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. *arXiv:2201.11903 [cs.CL]* <https://arxiv.org/abs/2201.11903>
- [99] Adam Wojciechowski, Mateusz Lango, and Ondrej Dusek. 2024. Faithful and Plausible Natural Language Explanations for Image Classification: A Pipeline Approach. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 2340–2351. <https://doi.org/10.18653/v1/2024.findings-emnlp.130>
- [100] Zhang Xi-Jia, Yue Guo, Shufei Chen, Simon Stepputtis, Matthew Gombolay, Katia Sycara, and Joseph Campbell. 2025. Model-agnostic policy explanations with large language models. *arXiv preprint arXiv:2504.05625* (2025).
- [101] Kelvin Xu, Jimmy Ba, Ryan Kiros, Kyunghyun Cho, Aaron Courville, Ruslan Salakhudinov, Rich Zemel, and Yoshua Bengio. 2015. Show, Attend and Tell: Neural Image Caption Generation with Visual Attention. In *Proceedings of the 32nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 37)*, Francis Bach and David Blei (Eds.). PMLR, Lille, France, 2048–2057. <https://proceedings.mlr.press/v37/xuc15.html>
- [102] Guang Yang, Qinghao Ye, and Jun Xia. 2022. Unbox the black-box for the medical explainable AI via multi-modal and multi-centre data fusion: A mini-review, two showcases and beyond. *Information Fusion* 77 (2022), 29–52.

- [103] Bingsheng Yao, Prithviraj Sen, Lucian Popa, James Hendler, and Dakuo Wang. 2023. Are Human Explanations Always Helpful? Towards Objective Evaluation of Human Natural Language Explanations. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 14698–14713.
- [104] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. arXiv:2305.10601 [cs.CL] <https://arxiv.org/abs/2305.10601>
- [105] I-Cheng Yeh. 2009. Default of Credit Card Clients. UCI Machine Learning Repository. DOI: <https://doi.org/10.24432/C55S3H>.
- [106] Zhitao Ying, Dylan Bourgeois, Jiaxuan You, Marinka Zitnik, and Jure Leskovec. 2019. Gnnexplainer: Generating explanations for graph neural networks. *Advances in neural information processing systems* 32 (2019).
- [107] Peipeng Yu, Jianwei Fei, Hui Gao, Xuan Feng, Zhihua Xia, and Chip Hong Chang. 2025. Unlocking the Capabilities of Large Vision-Language Models for Generalizable and Explainable Deepfake Detection. *arXiv preprint arXiv:2503.14853* (2025).
- [108] Peipeng Yu, Jianwei Fei, Hui Gao, Xuan Feng, Zhihua Xia, and Chip Hong Chang. 2025. Unlocking the Capabilities of Large Vision-Language Models for Generalizable and Explainable Deepfake Detection. In *Forty-second International Conference on Machine Learning*.
- [109] Jeffrey C Zemla, Steven Sloman, Christos Bechlivanidis, and David A Lagnado. 2017. Evaluating everyday explanations. *Psychonomic bulletin & review* 24, 5 (2017), 1488–1500.
- [110] Xianlong Zeng. 2024. Enhancing the Interpretability of SHAP Values Using Large Language Models. arXiv:2409.00079 [cs.HC] <https://arxiv.org/abs/2409.00079>
- [111] Jiaxing Zhang, Zhuomin Chen, Longchao Da, Dongsheng Luo, Hua Wei, et al. 2024. Regexplainer: Generating explanations for graph neural networks in regression tasks. *Advances in Neural Information Processing Systems* 37 (2024), 79282–79306.
- [112] Banghua Zhu, Evan Frick, Tianhao Wu, Hanlin Zhu, Karthik Ganesan, Wei-Lin Chiang, Jian Zhang, and Jiantao Jiao. 2024. Starling-7b: Improving helpfulness and harmlessness with rlaiif. In *First Conference on Language Modeling*.
- [113] Alexandra Zyttek, Sara Pido, Sarah Alnegheimish, Laure Berti-Equille, and Kalyan Veeramachaneni. 2024. Explingo: Explaining ai predictions using large language models. In *2024 IEEE International Conference on Big Data (BigData)*. IEEE, 1197–1208.
- [114] Alexandra Zyttek, Sara Pidò, and Kalyan Veeramachaneni. 2024. Llms for xai: Future directions for explaining explanations. *arXiv preprint arXiv:2405.06064* (2024).